

The New Hadley Centre Climate Model (HadGEM1): Evaluation of Coupled Simulations

T. C. JOHNS,* C. F. DURMAN,* H. T. BANKS,* M. J. ROBERTS,* A. J. McLAREN,* J. K. RIDLEY,*
C. A. SENIOR,* K. D. WILLIAMS,* A. JONES,* G. J. RICKARD,⁺ S. CUSACK,* W. J. INGRAM,[#]
M. CRUCIFIX,* D. M. H. SEXTON,* M. M. JOSHI,* B.-W. DONG,* H. SPENCER,[@] R. S. R. HILL,*
J. M. GREGORY,[&] A. B. KEEN,* A. K. PARDAENS,* J. A. LOWE,* A. BODAS-SALCEDO,* S. STARK,*
AND Y. SEARL*

**Met Office, Exeter, United Kingdom*

+NIWA, Wellington, New Zealand

[#]Met Office, Exeter, and Department of Physics, University of Oxford, Oxford, United Kingdom

@CGAM, University of Reading, Reading, United Kingdom

& Met Office, Exeter, and CGAM, University of Reading, Reading, United Kingdom

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ABSTRACT

A new coupled general circulation climate model developed at the Met Office's Hadley Centre is presented, and aspects of its performance in climate simulations run for the Intergovernmental Panel on Climate Change Fourth Assessment Report (IPCC AR4) documented with reference to previous models. The Hadley Centre Global Environmental Model version 1 (HadGEM1) is built around a new atmospheric dynamical core; uses higher resolution than the previous Hadley Centre model, HadCM3; and contains several improvements in its formulation including interactive atmospheric aerosols (sulphate, black carbon, biomass burning, and sea salt) plus their direct and indirect effects. The ocean component also has higher resolution and incorporates a sea ice component more advanced than HadCM3 in terms of both dynamics and thermodynamics. HadGEM1 thus permits experiments including some interactive processes not feasible with HadCM3. The simulation of present-day mean climate in HadGEM1 is significantly better overall in comparison to HadCM3, although some deficiencies exist in the simulation of tropical climate and El Niño variability. We quantify the overall improvement using a quasi-objective climate index encompassing a range of atmospheric, oceanic, and sea ice variables. It arises partly from higher resolution but also from greater fidelity in modeling dynamical and physical processes, for example, in the representation of clouds and sea ice. HadGEM1 has a similar effective climate sensitivity (2.8 K) to a CO₂ doubling as HadCM3 (3.1 K), although there are significant regional differences in their response patterns, especially in the Tropics. HadGEM1 is anticipated to be used as the basis both for higher-resolution and higher-complexity Earth System studies in the near future.

1. Introduction

In this paper, we present the Hadley Centre Global Environmental Model version 1 (HadGEM1), a new climate model developed at the Hadley Centre, and describe some aspects of its performance. HadGEM1 is intended as a platform for incorporating components of the environmental system other than just physical climate.

The initial motivation for the development of HadGEM1 was to achieve better scientific performance than the previous Hadley Centre model, HadCM3 (Pope et al. 2000; Gordon et al. 2000), delivering higher physical resolution and greater experimental flexibility. HadGEM1 has at its center a new nonhydrostatic atmospheric dynamical core ["New Dynamics" (ND), Davies et al. (2005)], employing a semi-implicit, semi-Lagrangian time integration scheme that is also used in the operational Numerical Weather Prediction (NWP) System of the United Kingdom Meteorological Office (Met Office). The atmospheric physical parameterization schemes in HadGEM1 have been improved considerably and remain closely in step with the NWP

Corresponding author address: Timothy C. Johns, Hadley Centre, Met Office, FitzRoy Road, Exeter EX1 3PB, United Kingdom.

E-mail: tim.johns@metoffice.gov.uk

model, thereby continuing the Unified Model (UM) strategy for NWP and climate modeling at the Met Office. HadGEM1 is also designed from the outset as a potential community climate model for U.K. university scientists and to be suitable for further development toward an Earth System modeling capability, that is, enabling coupling of physical climate with chemistry and ecosystem subcomponents. However, HadGEM1 as described herein remains a conventional atmosphere–ocean general circulation model (AOGCM) but incorporating significant scientific changes to HadCM3, particularly with regard to its atmospheric and sea ice components.

We built on established project and team-based model development approaches and, similarly to other centers (e.g., GFDL Global Atmospheric Model Development Team 2004), drew on scientific expertise spread across a large team—both within the Met Office and via external scientific contacts. Scientific performance issues identified in development were tackled by expert teams, working over periods of a year or so and considering both climate and NWP model configurations, with successive improvements being consolidated into the HadGEM1 prototype. The final stage of development explored parameter-based tuning to optimize the skill of the model while keeping any imbalances and implied drifts in the coupled model within an acceptable tolerance.

It is beyond the scope of this paper to provide a full scientific description of HadGEM1, but in section 2 we briefly describe the main model components and some factors influencing the model design (other papers and technical reports referenced therein provide more details). In section 3 we outline the experiments performed with HadGEM1. In section 4 we discuss some key criteria used to determine acceptable model performance and quantify HadGEM1 “skill” compared to HadCM3 using a quasi-objective statistical method. In sections 5–7 we evaluate aspects of HadGEM1’s control simulation in more detail, drawing comparisons both with observed climatological fields and HadCM3 (discussing areas of both improvement and degradation in model skill). In section 8 we attempt to explain changes in climate sensitivity and local feedbacks between HadCM3 and HadGEM1. Section 9 is our concluding discussion.

2. Model description

The starting points for building HadGEM1 were the ND dynamical core (Davies et al. 2005) plus the physical and technical improvements made with respect to HadCM3 in two intermediate models: the atmospheric

component, the Hadley Centre Atmospheric Model version 4 (HadAM4) (Webb et al. 2001), and the coupled model, the Hadley Centre Coupled Eddy-permitting Model (HadCEM) (Roberts et al. 2004). The atmospheric physics improvements in HadAM4 were developed within HadCM3’s old dynamical core however, so building the atmospheric component of HadGEM1 around ND necessitated recoding of the HadAM4 physics schemes to interface them correctly.

a. Computational cost and grid resolution

HadCM3 (Pope et al. 2000; Gordon et al. 2000) became operational late in 1997. Projecting a normal rate of increase in computer power up to 2003/04 (the target for operational delivery of HadGEM1) implied that a model having approximately 15–20 times the computational cost of HadCM3 would be affordable by that time. This was factored in as roughly a 6–8 times increase from higher resolution, and 2–3 times from additional physical complexity.

The standard atmospheric component of HadGEM1 (HadGAM1 N96L38) discussed here uses a horizontal resolution of $1.25^\circ \times 1.875^\circ$ in latitude and longitude with 38 layers in the vertical extending to over 39 km in height, given in Table 2 of Martin et al. (2006). (“N96” is our shorthand denoting a resolution of 96 two-grid-length waves, that is, 192 grid points in longitude.) The oceanic component of HadGEM1 (HadGOM1) uses a latitude–longitude grid with a zonal resolution of 1° everywhere and meridional resolution of 1° between the poles and 30° latitude, from which it increases smoothly to $\frac{1}{3}^\circ$ at the equator, giving 360×216 grid points in total. It has 40 unevenly spaced levels in the vertical reducing to 10-m thickness near the surface (Table 1).

HadGEM1 has 8 times as many atmosphere grid points as HadCM3 ($2 \times 2 \times 2$ in latitude, longitude, and vertical dimensions) but only 3.75 times as many oceanic grid points ($1.5 \times 1.25 \times 2$ in latitude, longitude, and depth). As neither the atmosphere nor ocean time steps differ from HadCM3, the total cost increase is around 6 times due to resolution factors alone. Other changes to physics and dynamics account for an additional factor estimated at 2.5 times, giving an overall cost increase in HadGEM1 of approximately 15 times compared to HadCM3, consistent with the original design. (This factor applies to our NEC SX6 supercomputer but may depend on the technical implementations and optimizations applied.)

The fact that ocean and atmosphere resolutions are more similar in HadGEM1 and higher than in HadCM3 is likely to improve coupling in that more spatial detail is contained in the fields forcing the ocean.

TABLE 1. Depth and thickness of ocean model levels (centered on tracer points).

Level	Depth (m)	Thickness (m)	Level	Depth (m)	Thickness (m)
1	5.0	10.0	21	398.60	80.8
2	15.0	10.0	22	488.25	98.5
3	25.0	10.0	23	596.40	117.8
4	35.0	10.0	24	724.50	138.4
5	45.0	10.0	25	873.85	160.3
6	55.0	10.0	26	1045.60	183.2
7	65.0	10.0	27	1240.70	207.0
8	75.0	10.0	28	1459.90	231.4
9	85.0	10.0	29	1703.70	256.2
10	95.0	10.0	30	1972.40	281.2
11	105.0	10.0	31	2265.80	305.6
12	115.0	10.0	32	2581.75	326.3
13	125.0	10.0	33	2914.95	340.1
14	135.6	11.2	34	3257.50	345.0
15	148.5	14.6	35	3602.50	345.0
16	165.95	20.3	36	3947.5	345.0
17	190.25	28.3	37	4292.5	345.0
18	223.6	38.4	38	4637.5	345.0
19	268.1	50.6	39	4982.5	345.0
20	325.8	64.8	40	5327.5	345.0

b. Atmosphere and land surface component

HadGAM1, the atmospheric (and land surface) component of HadGEM1, differs markedly from the predecessor model HadAM3, the most significant structural difference being the move to the ND core.

The advances offered by ND, expanded on by Davies et al. (2005), include

- a nonhydrostatic, fully compressible, deep atmosphere formulation with fewer approximations to the basic equations;
- a semi-implicit semi-Lagrangian time integration scheme for advection of all prognostic variables except density (for which a semi-implicit but Eulerian scheme is used), permitting relatively long time steps to be used at high resolution;
- better geostrophic adjustment properties, bringing better balance (and reduced time step dependence) to the coupling with physical parameterizations, which now calculate increments in parallel based on balanced states rather than in sequence (Davies et al. 2005);
- a conservative and monotone treatment of tracer transport.

The benefit of a nonhydrostatic formulation for a model resolution $O(150\text{ km})$ is arguable, but ND opens up the possibility of running at much higher resolution in future (down to 5 km or lower) without further major revisions to the dynamics (though revisions to the

physical schemes at such scales are much more likely to be necessary).

Another key advance in HadGAM1 is the inclusion as a standard feature of the interactive modeling of atmospheric aerosols, driven by surface and elevated emissions (from both natural and anthropogenic sources), including tropospheric chemical processes as well as physical removal processes such as washout. This scheme is coupled to the atmospheric model dynamics and physics, including radiation, cloud microphysics, precipitation, and boundary layer, the aerosol species represented being sulphate, black carbon, biomass smoke, and sea salt (Jones et al. 2001; Woodage et al. 2003; Davison et al. 2004; Roberts and Jones 2004). The coupling occurs on each time step (3-hourly for radiative calculations, every 30 min for other dynamics and physics). Modeling atmospheric aerosols removes the need for a climatological aerosol representation, which was included in HadCM3. It also permits modeling of the direct and indirect effects of aerosols, which are important forcings and major sources of modeling uncertainty under climate change (Ramaswamy et al. 2001). Note that HadCM3 modeled the direct effect of sulphate aerosol, but generally only that due to anthropogenic emissions (e.g., Johns et al. 2003) rather than from combined natural and anthropogenic emissions.

There are many other substantial differences in the physical parameterizations in HadGAM1 compared to HadAM3 that are beyond the scope of this paper to discuss in detail. The more important new features in HadGAM1 are

- a revised function for the sea surface albedo and introduction of a spectral dependence;
- a scheme to make the vertical ozone profile track the dynamically changing tropopause;
- a major upgrade to version II of the Met Office Surface Exchange Scheme (MOSES-II: Cox et al. 1999; Essery et al. 2003) that includes a land-surface-type tiling scheme and land-sea tiling facilitating flux-conserving coupling to the ocean grid;
- the introduction of a dynamical river transport model [Total Runoff Integrating Pathways (TRIP), Oki and Sud (1998)];
- a substantially modified planetary boundary layer mixing scheme (Lock et al. 2000);
- a major upgrade to the mass flux convection scheme including explicit coupling to the planetary boundary layer scheme, separate diagnosis of shallow and deep convection, revisions to the parameterization of convective momentum transport and its closure at cloud base, and a simple representation of the radiative effect of convective anvils;

- an interactive parameterization of the grid box critical relative humidity (determining the cloud water and cloud amount) based on local variability in the humidity field;
- the addition of a prognostic cloud ice variable and an upgraded microphysics scheme (based on Wilson and Ballard 1999), determining transfers between ice, liquid water, water vapor, and rain categories;
- modifications to the gravity wave drag scheme, including a representation of low-level flow blocking (Webster et al. 2003).

Martin et al. (2006) discuss these differences further and analyze their impact on the atmospheric performance of HadGEM1.

c. Ocean component

The ocean component HadGOM1 is based on the Bryan–Cox code (Bryan 1969; Cox 1984), as was the ocean component of HadCM3 (Gordon et al. 2000). It uses a bathymetry derived from the Smith and Sandwell (1997) $\frac{1}{30}^\circ$ depth dataset merged with ETOPO5 (1988) $\frac{1}{12}^\circ$ data at high latitudes, interpolated to the model grid and smoothed using a five-point (1:4:1) two-dimensional filter. Where this procedure obstructs important narrow pathways (e.g., Denmark Strait, Faroes–Shetland Channel, Vema Channel, and around the Indonesian archipelago), the bathymetry is adjusted to allow some flow at realistic depths (with reference to Thompson 1995). We interpret this procedure as providing the most appropriate bathymetry for the given model resolution, rather than as a tuning exercise.

In brief, the main scientific differences in HadGOM1 compared to the ocean component of HadCM3 are as follows:

- The higher horizontal and vertical resolution, land–sea mask, and bathymetry as previously mentioned. The increased horizontal resolution and stretched grid toward the equator, in conjunction with the new coupling scheme mentioned later, allow features to be captured well beyond the capability of HadCM3. For example, observations show that most of the Indonesian Throughflow passes through Makassar Strait (Gordon and Fine 1996). In HadCM3 this strait is blocked (Banks 2000), whereas the enhanced resolution of HadGEM1 allows the bathymetry and currents in this region to be represented in more detail (Fig. 1).
- An implicit linear free surface scheme based on Dukowicz and Smith (1994) with explicit freshwater fluxes to allow a more realistic representation of freshwater forcing. The fixed reference salinity value

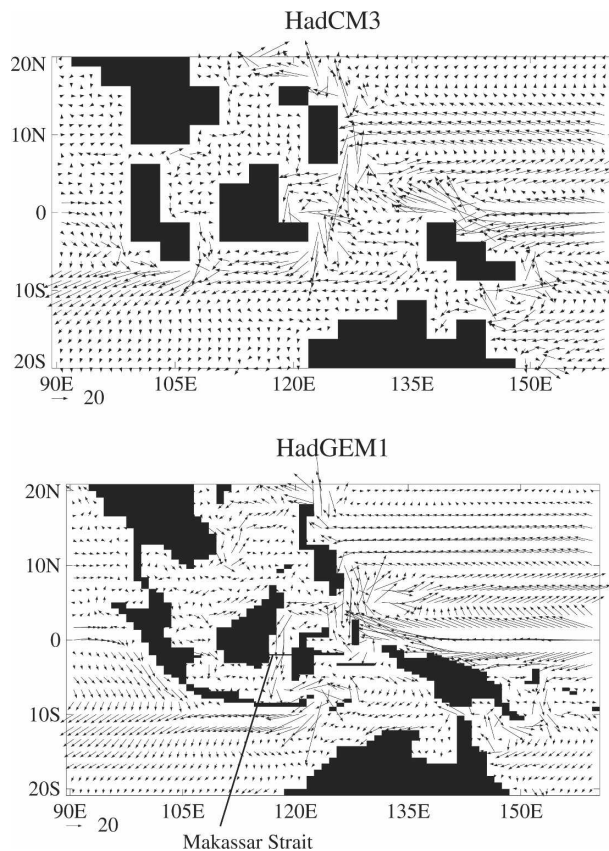


FIG. 1. Land and sea distributions in the vicinity of the Maritime Continent and ocean currents (cm s^{-1}) at 15-m depth as represented in (top) HadCM3 and (bottom) HadGEM1.

used in HadCM3 to convert freshwater fluxes into a “virtual salt flux” is now not used.

- A pseudo fourth-order advection scheme (Pacanowski and Griffies 1998), which is more accurate and generates less grid-scale noise than HadCM3’s second-order scheme, including upstream mixing at the ocean bottom to improve model stability there.
- A simple semi-implicit representation of linear bottom friction.
- Isopycnal diffusivity takes a constant value of $500 \text{ m}^2 \text{ s}^{-1}$ using the Griffies et al. (1998) scheme; the Gent and McWilliams (1990) adiabatic mixing scheme in the skew flux form (Griffies 1998) is used with a spatially and temporally varying coefficient (Visbeck et al. 1997; Roberts 2004), a minimum value of $150 \text{ m}^2 \text{ s}^{-1}$, and spatial distribution with higher values in the western boundary currents and Antarctic Circumpolar Current. The biharmonic adiabatic scheme of Roberts and Marshall (1998) is used with coefficient $1.0 \times 10^{12} \cos^3(\text{lat}) \text{ m}^4 \text{ s}^{-1}$. The $\cos(\text{lat})$ factor is required for numerical reasons owing to the convergence of meridians at high latitude.

- Horizontally aligned biharmonic tracer mixing in the top 20 m (top two model layers), with coefficient $2.5 \times 10^{12} \cos^3(\text{lat}) \text{ m}^4 \text{ s}^{-1}$, to partially represent enhanced mixing at the ocean surface.
- Momentum diffusion uses both a Laplacian scheme with a constant coefficient of $2000 \text{ m}^2 \text{ s}^{-1}$ and a biharmonic scheme with coefficient $1.0 \times 10^{13} \cos^3(\text{lat}) \text{ m}^4 \text{ s}^{-1}$.
- Changes to vertical mixing schemes. Temperature and salinity profiles are determined by a simple Kraus–Turner mixed layer scheme (Kraus and Turner 1967) in which work done against gravity in mixing down buoyant water is balanced by turbulent energy input from the wind. This scheme determines the tracer and density mixed layer depth but not the vertical current profile. Observations show that the currents are often not well mixed and that the wind-driven ageostrophic current often penetrates well below the mixed layer (e.g., Wijffels et al. 1994 and references therein). To represent the mixing of momentum in the mixed layer we assume the diffusion coefficient K_m to be a quadratic function of depth as in HadCM3 (see also Large et al. 1994). Mixed layer models often produce gradients at the mixed layer base that are too sharp, so an additional modification is included to enhance mixing in the transition region immediately below the mixed layer. Some modifications are also made to parameters in the Peters et al. (1988) Richardson-number-dependent vertical mixing scheme, applied beneath the mixed layer, to reduce noise.
- The Kraus–Turner bulk mixed layer now uses a value of $\lambda = 0.55$ for the scale factor determining the wind mixing energy available for overturning stable stratification (defined as $\lambda \rho_w u_*^3$, where ρ_w is the density of water and u_* the surface friction velocity). This gives a better fit to data from Argo and Ocean Weather Ship “Papa” than the value of $\lambda = 0.7$ used in HadCM3 (Acreman 2005).
- The Roussinov convective adjustment scheme (Rother et al. 1994) is not used.
- Flow exchanged with three marginal seas whose connections with the World Ocean are not resolved by the model grid is parameterized. The Mediterranean Sea has 0.4 Sv ($\text{Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$) of water fluxed in over the top 80 m and out at 600 m, the Red Sea has 0.2 Sv fluxed in over the top 20 m and out at 40–60 m, and the Persian Gulf has 0.1 Sv fluxed in over the top 20 m and out at 40–60 m.

d. Sea ice component

Sea ice plays an important role in global climate due to its high surface albedo, insulating effect on the

ocean, and influence on the ocean salinity through brine rejection when ice forms and surface freshening when ice melts. Hence it is important to produce a realistic simulation of sea ice within the coupled model.

The sea ice component of HadGEM1 is more complex than HadCM3, with improvements made to both the thermodynamics and dynamics by incorporating components of the Los Alamos National Laboratory sea ice model (CICE) (Hunke and Lipscomb 2004). HadGEM1 resolves the subgrid-scale ice thickness distribution (ITD), its evolution being determined by thermodynamic growth/melt, advection, and redistribution by ridging (Thorndike et al. 1975).

The differences in the HadGEM1 sea ice model relative to HadCM3 are:

- The ice pack is divided into five thickness categories and open water, a choice that is sufficient to resolve the ITD for climate modeling (Bitz et al. 2001; Lipscomb 2001). (The ice pack in HadCM3 was represented by a single ice thickness and open water in each grid box.)
- The linear remapping scheme of Lipscomb (2001) is used to compute the thermodynamic transfer of ice between categories.
- Ice velocities are calculated using elastic–viscous–plastic (EVP) ice dynamics (Hunke and Dukowicz 1997), whereas in HadCM3 the ice was advected with the surface ocean currents.
- HadGEM1 uses the CICE model ice ridging scheme (Hunke and Lipscomb 2004).
- Ice and snow albedos respond separately to the surface temperature, with partial snow cover providing a modification to the bare ice albedo similar to the scheme of Cox et al. (1999).

The thermodynamic growth/melt of each ice category is modeled using the zero-layer model of Semtner (1976; as in HadCM3). For a more comprehensive description of the sea ice scheme, see McLaren et al. (2005, unpublished manuscript, hereafter MCL), which also states the main sea ice parameter values.

e. Coupling between atmosphere, land surface, ocean, and sea ice

HadGAM1 is physically coupled to HadGOM1 using a daily cycle that passes mean daily surface atmosphere-to-ocean (and sea ice) fluxes (wind stress, penetrative solar radiation, nonpenetrative net heat flux, precipitation minus evaporation, river outflow, snowfall, sublimation, and sea ice top and bottom melting) and ocean surface boundary conditions (surface cur-

rent, ice concentration, ice depth, snow depth on ice, and sea surface temperature—fixed for the atmosphere day) to the atmosphere. The coupling fields are spatially two-dimensional apart from the sea ice properties, and top and bottom melting fluxes that have an extra dimension representing ice thickness categories (see appendix A of Johns et al. 2004).

The land–sea masks for HadGAM1 and HadGOM1 differ because of their different horizontal resolution. To enable conservative flux coupling between them, a coastal tiling method is applied within the MOSES-II surface exchange scheme (Essery et al. 2003). Fluxes are computed separately for land, ocean, and sea ice fractions of each atmosphere grid box (fractions determined from the fixed geometry of the atmosphere grid overlying the ocean model coastline plus the time-varying sea ice state), then passed to the appropriate model components such that the total flux is accounted for. The atmosphere thus experiences a fuzzy coastline (a mixture of land and sea surface types) rather than a sharp land–sea division as in HadCM3.

Coupling between the atmosphere and ocean consists of several steps. Fluxes are transferred to the ocean grid via horizontal bilinear interpolation with a correction for discontinuities in interpolating across the land–sea boundary. A local weighting is then applied to the interpolated fields to ensure that fluxes are globally conserved. The interpolation scheme follows that employed in HadCEM (Roberts et al. 2004).

Rivers are modeled by an embedded river routing scheme based on TRIP (Oki and Sud 1998), which includes river transport dynamics, driven by fluxes of surface and subsurface runoff. TRIP operates on a higher-resolution, $1^\circ \times 1^\circ$ grid, than the atmosphere and land surface model, necessitating additional coupling to transfer the runoff fluxes and integrated river flows, the latter being deposited at predefined coastal outflow points on the atmosphere grid then passed to the ocean model as a surface freshwater flux term. An error in the conservation of freshwater in this coupling to the ocean has been identified, which affects all the model results presented later, some river outflow at some grid boxes being lost to the system. This leads to an erroneous downward drift of global free surface height, but the effect on the simulation is otherwise minor (we checked this by comparing with a 5-yr parallel run with the bug fixed but, owing to computer time constraints, were unable to rerun the whole set of simulations).

As in HadCM3, the accumulation of frozen water on the permanent ice sheets is never returned to the freshwater cycle; that is, there is no representation of icebergs calving off ice shelves. Runoff draining into inland basins is also lost to the system. To counterbalance

these sinks in the global annual mean freshwater budget a freshwater flux field is applied to the ocean to add back a flux, invariant in time, with a pattern and scaling the same as that calibrated for HadCM3 but interpolated to the HadGOM1 grid. (Note that this does not adjust for the water conservation error mentioned above.)

f. Other model configurations

As well as running in atmosphere-only mode with prescribed surface forcing (Martin et al. 2006), HadGAM1 can also be coupled to a 50-m thermodynamic mixed layer ocean and sea ice (slab) model, forming HadGSM1. The sea ice dynamics and thermodynamics parameterizations in HadGSM1 are the same as in HadGEM1 except for the following differences in HadGSM1:

- Ocean to ice heat flux is calculated as a simple function of the difference between the SST and freezing point of seawater.
- Climatological ocean currents from the HadGEM1 control run are used as input to the EVP ice velocity calculation.
- The slab ocean has the same resolution as HadGAM1, hence coastal ocean points are tiled. For numerical reasons, sea ice at such points can only change thermodynamically, not dynamically.

The ocean surface temperature is maintained close to climatological values by use of a monthly varying heat flux field calculated in a calibration experiment in which SSTs are reset to climatological values on each time step.

3. Experimental design and initialization

A number of transient HadGEM1 runs have been conducted, with the Intergovernmental Panel on Climate Change Fourth Assessment Report (IPCC AR4) particularly in mind, as described by Stott et al. (2006, hereafter S06).

In this paper we focus on the coupled HadGEM1 control run with fixed 1860 forcing levels for greenhouse gases, ozone, sulphur, and other aerosol precursor emissions and land surface boundary conditions. No sporadic volcanic eruptions are included. This run was initialized and spun up from an ocean state, using climatological fields for September mean potential temperature and salinity (Levitus et al. 1998), at rest. These were interpolated to the defined model levels and bathymetry with various infilling and adjustment procedures to avoid obvious instabilities or inconsistency.

The sea ice was initialized as described by MCL and the atmosphere from an analyzed state corresponding to 1 September 1978, also used for initializing the atmosphere-only Atmospheric Model Intercomparison Project II (AMIP-II) runs with HadGAM1 described by Martin et al. (2006).

Note that a few scientific changes were made early in the spinup phase:

- To correct an inconsistency between the effective Fourier filtering of modes for the barotropic and baroclinic ocean velocities (minor impact).
- To correct an error in the albedo of snow-covered sea ice (affecting mainly the net radiation and to some extent the sea ice extents).
- To apply a convective available potential energy closure with a time scale dependent on relative humidity in the convecting column, already used for deep convection, to midlevel convection. This helps to limit the occurrence of numerical instabilities associated with grid-scale convection characterized by strong vertical motion.

Toward the end of the spinup a further change was made to correct an error in the convective momentum transport scheme that has an impact, though not a major one, mainly in the tropical warm pool and surrounding regions.

We chose to introduce 1860 forcing conditions immediately after initializing, although the initial states are not consistent with 1860 radiative forcing conditions, and continued with 1860 forcing through both spinup and control phases. A more consistent, but computationally intensive, method of initializing transient climate change experiments proposed by Stouffer et al. (2004)—that is, running the forcing backward in time from the present day before holding it stable at 1860 for several centuries—was not adopted as it would have delayed completion of the full experiment by about a year. We estimate the net forcing difference at 1860 relative to present-day conditions to be approximately -1 W m^{-2} in HadGEM1, comprising about $+1 \text{ W m}^{-2}$ due to aerosols and -2 W m^{-2} due to nonaerosol forcings.

The HadGEM1 control run forms the backbone for the other transient forced runs (described in more detail by S06) that were started after 85 years of coupled spinup. These transient runs include an idealized experiment with a standard 1%-per-annum increase in CO_2 for 80 years starting from the control CO_2 level [a Coupled Model Intercomparison Project (CMIP) experiment]. The length of the spinup conducted before introducing time-dependent forcings in these runs is shorter than ideal but was considered enough to allow

the faster components of the climate system to approach equilibrium.

Control and equilibrium $2 \times \text{CO}_2$ simulations with the coupled slab model configuration of HadGEM1 (HadGSM1) have been conducted to assess the climate sensitivity and feedbacks in HadGEM1. These were run to equilibrium and then for a further 40 years (means over the last 40 yr being the basis for analysis).

A set of atmosphere-only runs using HadGAM1 forced with “AMIP-II” boundary conditions, running from 1978 to 2003 with corresponding greenhouse gas concentrations, also forms part of the experimental setup. An analysis of atmospheric performance both in these AMIP-II simulations and in the HadGEM1 control run is detailed by Martin et al. (2006), so we concentrate in the following sections predominantly on coupled aspects of HadGEM1 performance in the control simulation, which has run more than 400 years.

4. Criteria used to assess model skill

A number of broad criteria and methodologies were used to guide the development and tuning of HadGEM1 and in the final assessment of HadGEM1 skill in its control climate, three of which we outline next.

a. Global energy balance

An important prerequisite is that the control simulation should be stable with a close enough balance between shortwave and longwave fluxes at the top of the atmosphere (TOA) and small enough energy flux across the atmosphere–ocean–sea ice boundaries to prevent large global climate drifts and, preferably, with minimal trends in the ocean interior. The latter is particularly difficult to achieve, given a short spinup phase. HadGEM1 exhibits only a slight imbalance at the TOA, a fairly stable net heating of just over 0.3 W m^{-2} averaged over years 1–145 of the combined spinup/control, with a negligible trend in the control run subsequently. Although a larger imbalance than in HadCM3, which was very close to a balance throughout, this is small enough to be acceptable for multicentury time-scale experiments. (Note, however, that the TOA net heating is more consistent between HadGAM1 and HadGEM1 than was the case between HadCM3 and HadAM3.)

b. Thermohaline circulation

It remains challenging for nonflux-adjusted coupled models to simulate (and maintain) a realistic thermohaline circulation (THC), as this requires realism not just in the atmospheric heat and freshwater forcing of

the ocean but also the correct response from the ocean dynamics to the forcing, leading to maintenance of the underlying water masses and their density contrasts. Another of our prerequisites is that HadGEM1 should simulate the THC at least as well as HadCM3, but this was found to be very difficult to achieve unless N96 (or higher) atmospheric resolution was employed with ND. At the lower horizontal resolution of N48 ($=2.5^\circ \times 3.75^\circ$, as in HadCM3) the oceanic poleward heat transport implied in atmosphere-only and coupled simulations by the divergence of atmospheric heat fluxes was too weak, contributing to an unrealistic reduction of THC strength in coupled simulations at this resolution. Atmospheric storm tracks and eddy kinetic energy are much better simulated at N96 resolution than N48 with ND (Ringer et al. 2006), and this contributes to sustaining a more realistic THC, strongly influencing our choice of N96 resolution for HadGEM1.

c. "Climate prediction index"

Having first achieved a viable model in terms of the prerequisites outlined above, we attempt to define and measure the model skill more objectively to aid model tuning and to optimize the model's performance.

No universally accepted objective method of defining the overall skill of a coupled model in reproducing observed climate exists. Though it is arguably desirable and useful to have such an objective measure, the practicalities of tuning a model to be skillful across all likely applications mean that metrics used in such statistical skill measurements will contain an element of prior judgment in the choice of variables and weighting as well as the observations and error estimates adopted. The development and tuning phase of HadGEM1 used a simple weighted nondimensional index of root-mean-square errors compared to present-day climatological means for a selection of multiyear model mean fields (of monthly, seasonal, and annual data) to quantify improvements or otherwise in model performance (supported by more subjective examination of fields). These fields cover the atmosphere, land surface, air-sea fluxes, sea ice, and ocean and include some zonal mean and some scalar quantities. The statistical skill metric used is the "Climate Prediction Index" (CPI) [see Murphy et al. (2004) supplementary material for details of the method but note that the set of variables included in the CPI differs here from that used by Murphy et al. as it encompasses oceanic and more sea ice variables].

The CPI breakdown for the final version of HadGEM1 compared to HadCM3 (Fig. 2; in which low index values signify "good") reveals improvements in the majority of the elements, particularly cloud-related fields, but with some exceptions such as near-surface (1.5 m) tem-

perature and precipitation. It is clear that improvements in cloud are a major factor in the advancement of HadGEM1 skill, but other variables are essential contributory factors in this improved simulation of clouds. Taken overall, the weighted CPI score for HadGEM1 is considerably better than for HadCM3, and the two other aggregated "Taylor diagram" scores (Taylor 2001), namely correlation and ratio of standard deviations, are also improved (Table 2). The choice of weighting of the individual CPI elements has an effect on the magnitude of this improvement [we weight all elements equally with the exception of International Satellite Cloud Climatology Project (ISCCP) cloud types, which are weighted by one-third], but we consider it justified to assert that HadGEM1 possesses better overall skill in the simulation of mean climate than HadCM3.

We do not rely solely on the CPI results to decide the optimal choice of model tuning but also take into account more detailed (and subjective) evaluations of its constituent elements plus aspects of performance not captured by the CPI.

5. Atmospheric and surface climate

Martin et al. (2006) document the atmospheric performance of HadGEM1 both in coupled and forced (AMIP) experiments, and we refer the reader there for more details. In brief, the model generally performs very well, aided in particular by the doubled horizontal and vertical resolution compared to HadCM3, the more accurate dynamics, and improved physics-dynamics coupling, as well as substantial improvements in the physical parameterization schemes themselves. Most aspects of the simulated climate in terms of temperature, winds, and moisture in the free atmosphere are significantly better than HadCM3, as is the mean sea level pressure. Major improvements in both the representation of cloud and more consistency between the simulated top-of-atmosphere radiation budget and the cloud radiative forcing are achieved. Also of note are the improvements described by Martin et al. (2006) in the simulated transport of water vapor and tracers, the tropopause structure, and the surface pressure in the Arctic (which aids the sea ice simulation there).

Problem areas identified in HadGEM1 include precipitation and land temperature biases (which have deteriorated in some regions and seasons compared to HadCM3), latent heat flux, and surface wind stress biases in the Tropics. The Tropics is a region where the combined changes introduced in HadGEM1 result in near-surface errors with detrimental impacts on certain other aspects of the coupled model performance includ-

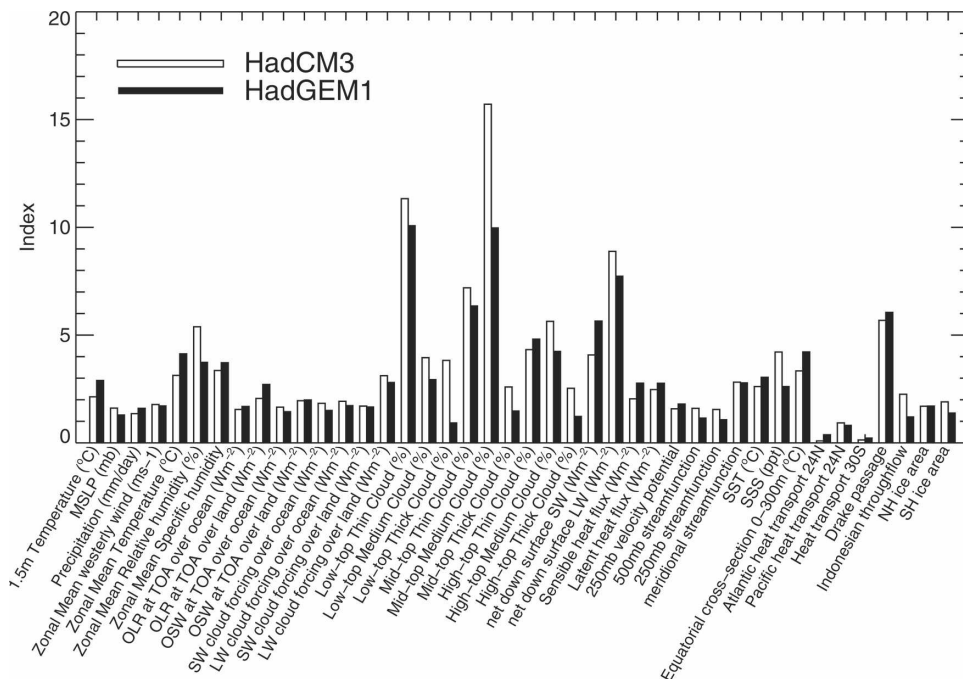


FIG. 2. Comparison of a nondimensional index of model skill compared with observed climatological fields between HadCM3 (open bars) and HadGEM1 (filled bars). Rms errors are normalized by the spatial average of internal climate variability estimated from HadCM3's control run for each variable shown, larger normalized rms errors being represented by longer bars. The index is similar to the CPI defined and used by Murphy et al. (2004) but contains more variables, including some oceanic and sea ice ones. The model data comprise averages of a 20-yr period early in the third century of the HadGEM1 control simulation (referenced to the start of the spinup) and a corresponding period of the HadCM3 control.

ing ENSO as described later—understanding and correcting this is a subject of future work.

6. Ocean and sea ice climate

a. Surface temperature and salinity errors

The model simulation of SST is extremely important as it is the primary mechanism for ocean–atmosphere coupling. A rapid adjustment occurs in the first decade

TABLE 2. CPI-weighted absolute statistics for HadGEM1 and HadCM3 (better score in bold). We have not adjusted the weightings of those individual elements in the original CPI as designed for other purposes (Murphy et al. 2004), but here we include additional elements. For each element, the rms error, spatial pattern correlation, and ratio of the standard deviations are computed with respect to corresponding climatological patterns following Taylor (2001), before averaging to form the weighted statistics shown.

Model	Nondimensional rms error (CPI)	Correlation	Ratio of std dev
HadCM3	3.054	0.834	1.499
HadGEM1	2.884	0.852	1.246

of the simulated SST fields as HadGEM1 develops an approximate balance between the atmosphere and ocean mixed layer. On a somewhat longer (multidecadal) time scale, the sea ice model approaches a stable seasonal cycle while SST and sea surface salinity (SSS) adjust further, reflecting the ocean gyre and overturning time scales. The SST and SSS errors, once established, show distinct and stable patterns that differ substantially from those exhibited by HadCM3 (Fig. 3). Note, however, that, if the global average bias is removed from the HadGEM1 error pattern, then, with the exception of the North Pacific, the HadGEM1 and HadCM3 error patterns match each other more closely.

The HadGEM1 SST error relative to HadISST is dominated by cold biases in the Tropics, subtropics, and northern midlatitudes. The large North Pacific cold bias in HadCM3 ($\sim 3^{\circ}\text{C}$) is reduced in HadGEM1 but the equatorial cold bias is larger. The warm bias exhibited by HadCM3 in the eastern subtropical ocean gyres is alleviated in HadGEM1 at least partly due to better representation of marine stratocumulus cloud (largely absent in HadCM3). This is a combined effect of higher vertical resolution, improved physics, and improved dy-

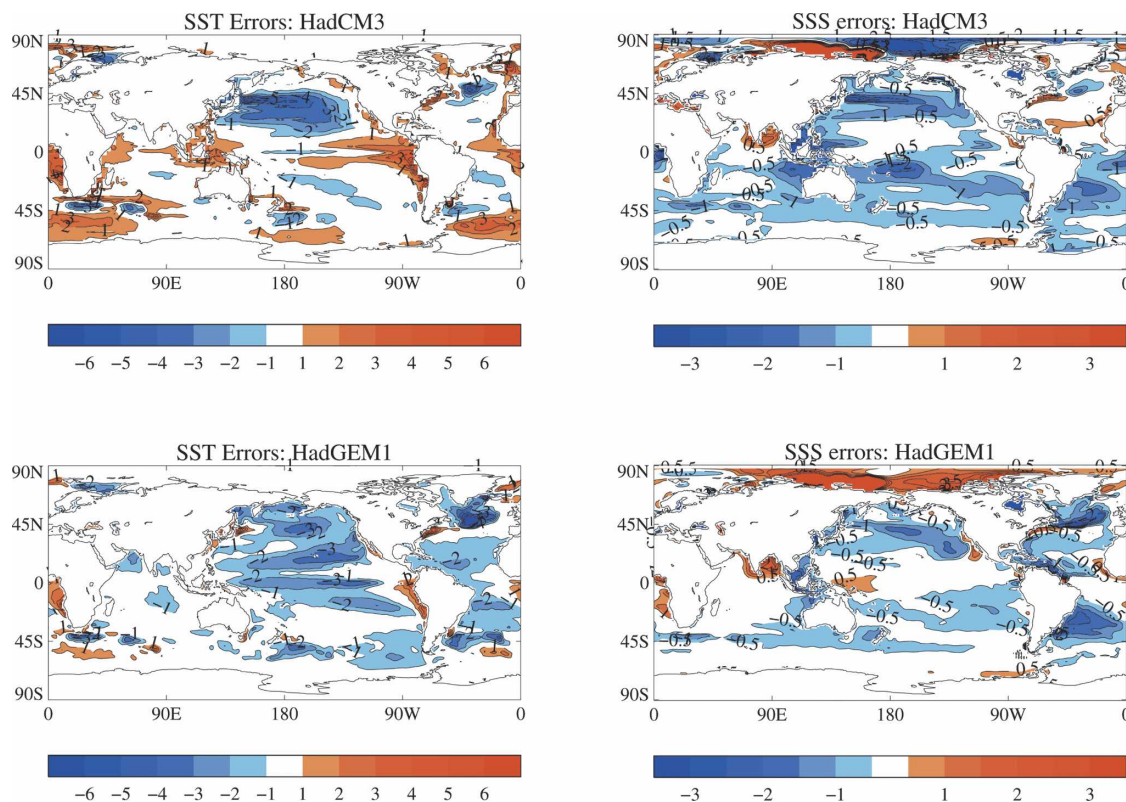


FIG. 3. Annual mean SST and SSS differences (error patterns), relative to HadISST (Rayner et al. 2003) for SST and Levitus (Levitus et al. 1998) for SSS, simulated by (top) HadCM3 and (bottom) HadGEM1. The model data compared are 20-yr averages for the same periods as in Fig. 2.

namics–physics coupling in the atmospheric component of HadGEM1, possibly assisted by the revised vertical grid staggering that now uses a Charney–Phillips grid [the properties and advantages of which are discussed by Davies et al. (2005) and Martin et al. (2006)]. The Southern Ocean warm bias in HadCM3 has also been removed in HadGEM1. While the global SST cold bias tends to improve local warm biases, changes in physical parameterizations are also likely to be contributing to this reduction in Southern Ocean SST errors. In sensitivity tests with HadCM3, the warm bias was reduced by a stronger ocean-to-ice heat flux coupling, including the effects of snow-on-ice albedo and a reduction in the along-isopycnal diffusion. All of these factors (noting that the new sea ice model in HadGEM1 parameterizes the ocean-to-ice heat flux differently to HadCM3) could be contributing to the improvements. In the northern Atlantic the cold SST errors are a little more extensive, at least partly due to a more zonal North Atlantic Current. Analysis of the large-scale cold SST bias patterns suggests that they are probably linked to excessively strong surface wind stresses and associated errors in heat fluxes, evaporation, and precipitation

(see section 6b), but the underlying reasons for these atmospheric biases, which are also present (though not as pronounced) in HadGAM1 AMIP-II experiments (Martin et al. 2006), are not yet understood.

The SSS errors in HadGEM1 relative to Levitus are generally an improvement on HadCM3, particularly in the Pacific and Indian Oceans (Fig. 3). The Atlantic fresh biases associated with the midlatitude gyres in the Northern and Southern Hemispheres are worse in HadGEM1 (the northern one, as with SST, being partly associated with a poorer simulation of the path of the North Atlantic Current in HadGEM1), while in the Arctic the bias is saline (but differences between climatologies are also large there). Some of the improvement in the surface salinity may be due to the explicit representation of freshwater fluxes in HadGEM1, as opposed to the virtual salt flux formulation of HadCM3.

b. Surface fluxes

Surface fluxes provide the external forcing for the thermodynamical and dynamical changes in the ocean. In this subsection we examine the surface fluxes from an ocean perspective. The net surface heat flux fields in

HadGEM1 and HadCM3 have been compared to both the unadjusted Southampton Oceanography Centre (SOC) climatology (Josey et al. 1996) and the da Silva climatology (da Silva et al. 1994). These differences are not shown as they are similar in both models. Of particular note in the differences between the two models (Fig. 4) is that in HadGEM1 there is a larger surface heat flux out of the ocean in the northwest Atlantic (consistent with the cold SST error in that region), and a larger net heat flux into the Southern Ocean.

The hydrological cycle in HadGEM1 is stronger than that in HadCM3 (globally averaged precipitation being about 0.15 mm day^{-1} higher in HadGEM1). Figure 5 shows that, relative to the Climate Prediction Center's Merged Analysis of Precipitation, Observation only (CMAP/O) climatology (Xie and Arkin 1997), HadGEM1 has too much precipitation over the Southern Ocean and the high latitudes of the North Atlantic and North Pacific. These errors are broadly similar to those exhibited by HadCM3 and, as was the case with HadCM3 (Pardaens et al. 2003), the results suggest that HadGEM1 has an overly strong hydrological cycle,

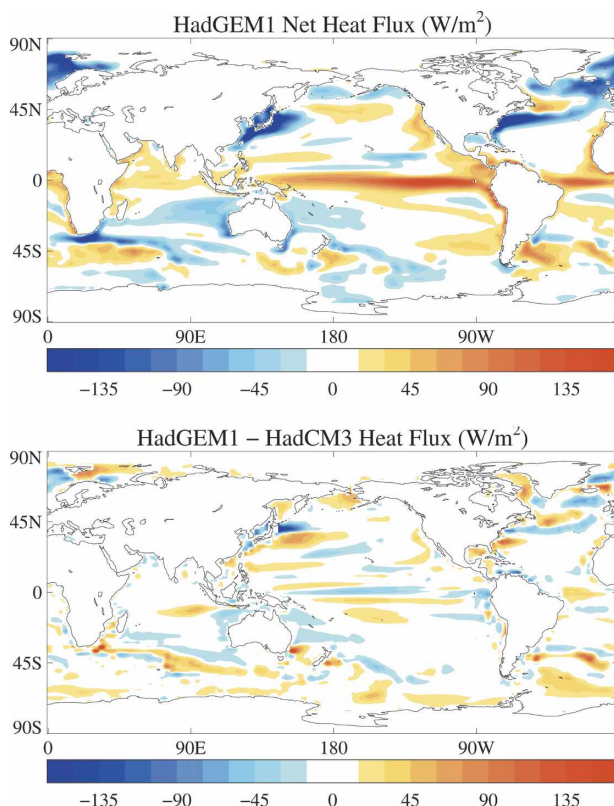


FIG. 4. Total heat flux in HadGEM1 and difference between HadGEM1 and HadCM3, for the same periods as in Fig. 2. The contour interval is 15 W m^{-2} with the range from -15 to $+15 \text{ W m}^{-2}$ left unshaded.

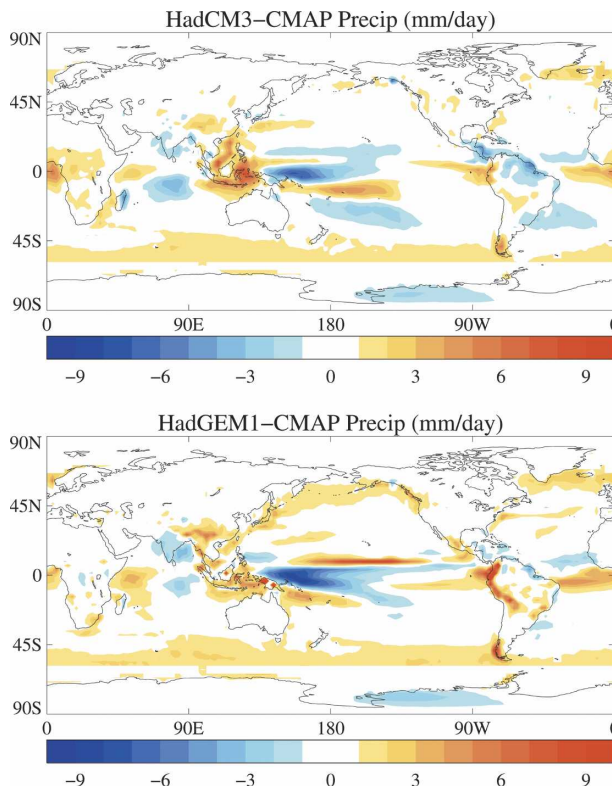


FIG. 5. Precipitation errors in HadGEM1 and HadCM3, for the same periods as in Fig. 2, compared to the CMAP/O climatology (Xie and Arkin 1997). The contour interval is 1 mm day^{-1} with the range from -1 to $+1 \text{ mm day}^{-1}$ left unshaded.

though the large uncertainties in the climatology make this difficult to quantify. A pronounced split intertropical convergence zone error, linked to the equatorial cold bias described earlier, is present in HadGEM1. Further details of the errors in HadGEM1 precipitation relative to climatology are given in Martin et al. (2006).

The wind stress in HadGEM1 is compared to the SOC climatology (Josey et al. 1996) in Fig. 6. Overall the model simulates the pattern of wind stress well, but the HadGEM1 stresses are stronger overall than in the climatology or in HadCM3. In particular, the westerly wind stress is stronger at high latitudes and the easterly winds are stronger over the subtropical gyres, with a larger meridional stress in the equatorial and tropical Pacific. The predominant error is over the Southern Ocean where the westerly wind stress is significantly overestimated compared to the climatology and significantly stronger in HadGEM1 than HadCM3, particularly in the Pacific sector. The lack of observations at these latitudes means that the SOC climatology is poorly constrained there, so, although the HadGEM1 wind stress is probably too strong, we cannot quantify this error precisely.

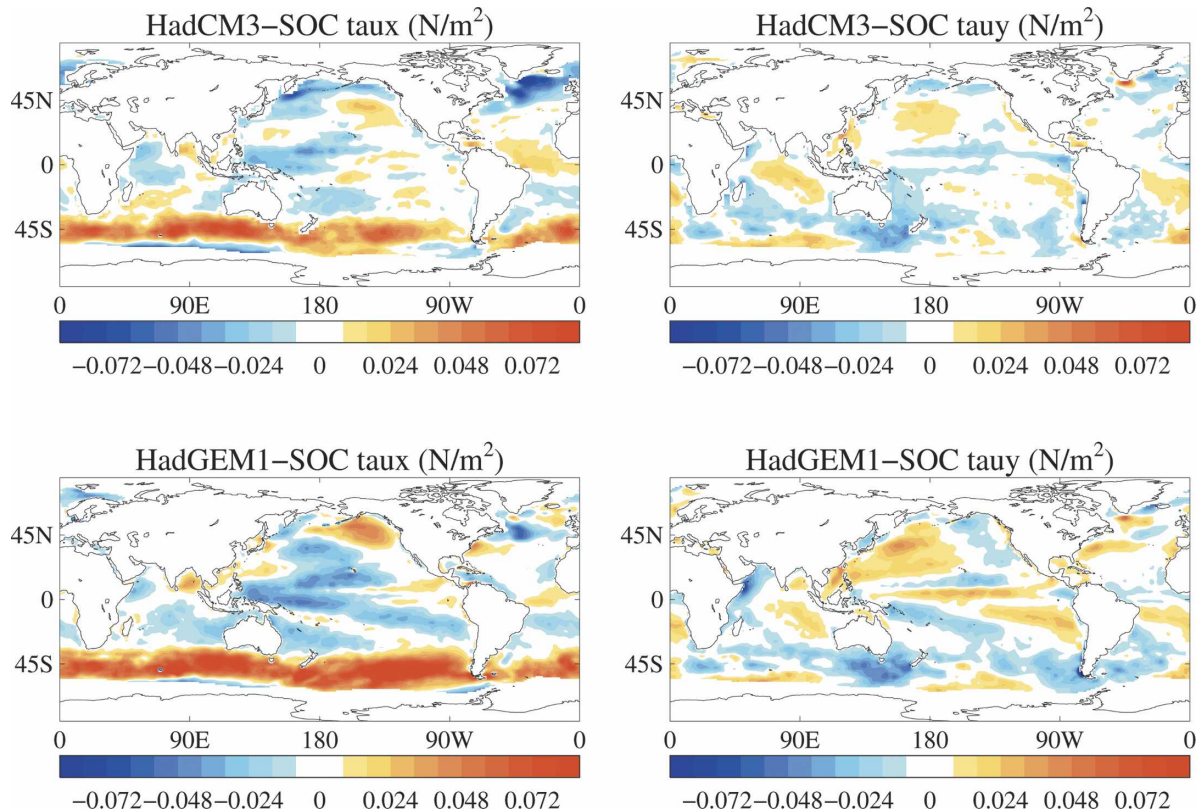


FIG. 6. Wind stress errors in HadGEM1 and HadCM3, for the same periods as in Fig. 2, compared to the SOC climatology (Josey et al. 1996). For each model the error in zonal stress (τ_x) and meridional stress (τ_y) is shown. The contour interval is 0.008 N m^{-2} with the range from -0.008 to $+0.008 \text{ N m}^{-2}$ left unshaded.

c. Subsurface temperature and salinity drifts

As discussed earlier, the net TOA radiation in HadGEM1 exhibits an imbalance (of $\sim 0.3 \text{ W m}^{-2}$ over the first 145 years) indicating a net warming of the climate system. As the heat capacities of the atmosphere and land are small, this residual energy must be taken up by the ocean. Examination of a time series of volume-weighted temperature reveals a net warming trend of 0.05°C per century, consistent with the heat gain indicated by the TOA flux. In contrast HadCM3 exhibited a cooling trend consistent with its generally small, negative TOA flux.

There will be a net drift in the volume-averaged potential temperature (salinity) if the ocean heat (freshwater) convergence does not balance the net surface heat (freshwater) flux terms. The global temperature and salinity drifts in the first four centuries of HadCM3 and HadGEM1 are shown in Fig. 7. HadGEM1 exhibits a rapid cooling in the upper 300 m that is not seen in HadCM3, as well as a stronger warming below this. In both models the ocean bottom waters cool slowly over time to a very similar degree—this signal will have a very long equilibration time scale (England 1995).

These HadGEM1 global drifts are similar to those seen in HadCEM (Roberts et al. 2004), which also had a positive TOA flux. Examining the drifts on a basin scale reveals that the largest contribution to the near-surface cooling is from the Pacific Ocean, while the largest contribution to the deeper warming is from the Atlantic Ocean.

The corresponding salinity drifts broadly indicate that both models freshen in the upper ocean and become more saline at depth, which may be consistent with an overly strong hydrological cycle (Pardaens et al. 2003). In both models the strongest global freshening trends are seen in the upper ocean, but this drift is larger in HadCM3. In HadCM3 deep Atlantic waters become very warm and saline through the first 400 years of the control run, while this trend is much weaker in HadGEM1, suggesting that some important water masses may be better represented in HadGEM1 than in HadCM3. Drifts in zonal mean Atlantic/Arctic salinity and density, with respect to the Levitus initial conditions (Fig. 8), are generally less in HadGEM1 than in HadCM3 after about 200 years, particularly in the deep northern Atlantic. This improvement is likely

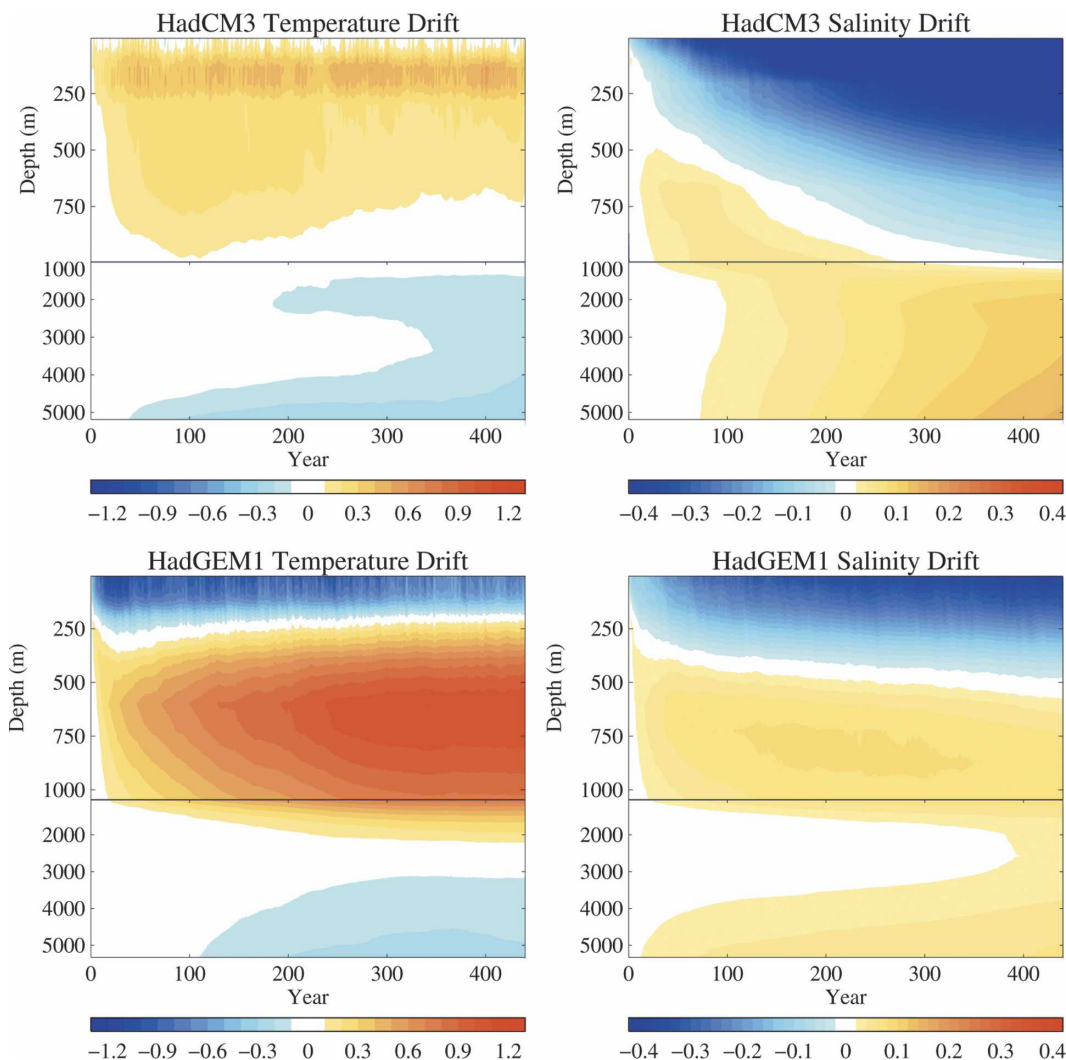


FIG. 7. Time series of global drifts in annual mean temperature ($^{\circ}\text{C}$) and salinity (psu) on level surfaces in (top) HadCM3 and (bottom) HadGEM1 relative to the first year of the simulations. For clarity the scale has been expanded (above the solid line) in the top 995 m for HadCM3 (top 12 model layers) and the top 1045 m for HadGEM1 (top 25 model layers).

to be important for simulating the thermohaline circulation correctly. Further analysis of the water mass structure in HadGEM1 will be a topic of future work.

d. Overturning and ocean heat transports

The strength of the North Atlantic Deep Water (NADW) cell in the upper 3000 m has a maximum of 18 Sv in HadGEM1 and 22 Sv in HadCM3, with this maximum shifted southward in HadGEM1 (Fig. 9). At 24°N , the observations of Hall and Bryden suggest a cell with strength 19.1 Sv, which is closer to the HadCM3 value (18 Sv) than HadGEM1 (15 Sv). The bottom cell associated with Antarctic Bottom Water transports around 5 Sv in HadGEM1, which is close to the 8 Sv of trans-

port observed through the Vema and Hunter Channels (Hogg et al. 1999; Zenk et al. 1999), while HadCM3 has an anomalous overturning cell of 16 Sv. The strength of the NADW overturning cell in HadCM3 is likely to be increased due to the influence of enhanced density of its overflows from the Greenland–Iceland–Norway Seas on the Atlantic meridional density gradient (Hughes and Weaver 1994). Although the net flux of water denser than 27.8 kg m^{-3} is comparable between the models [7.3 Sv for HadGEM1, 7.4 Sv for HadCM3, and 5.6 Sv observed by Dickson and Brown (1994)], in HadCM3 much of this water is at very high densities (greater than 28.2 kg m^{-3}).

Atlantic Ocean heat transports in HadCM3 and

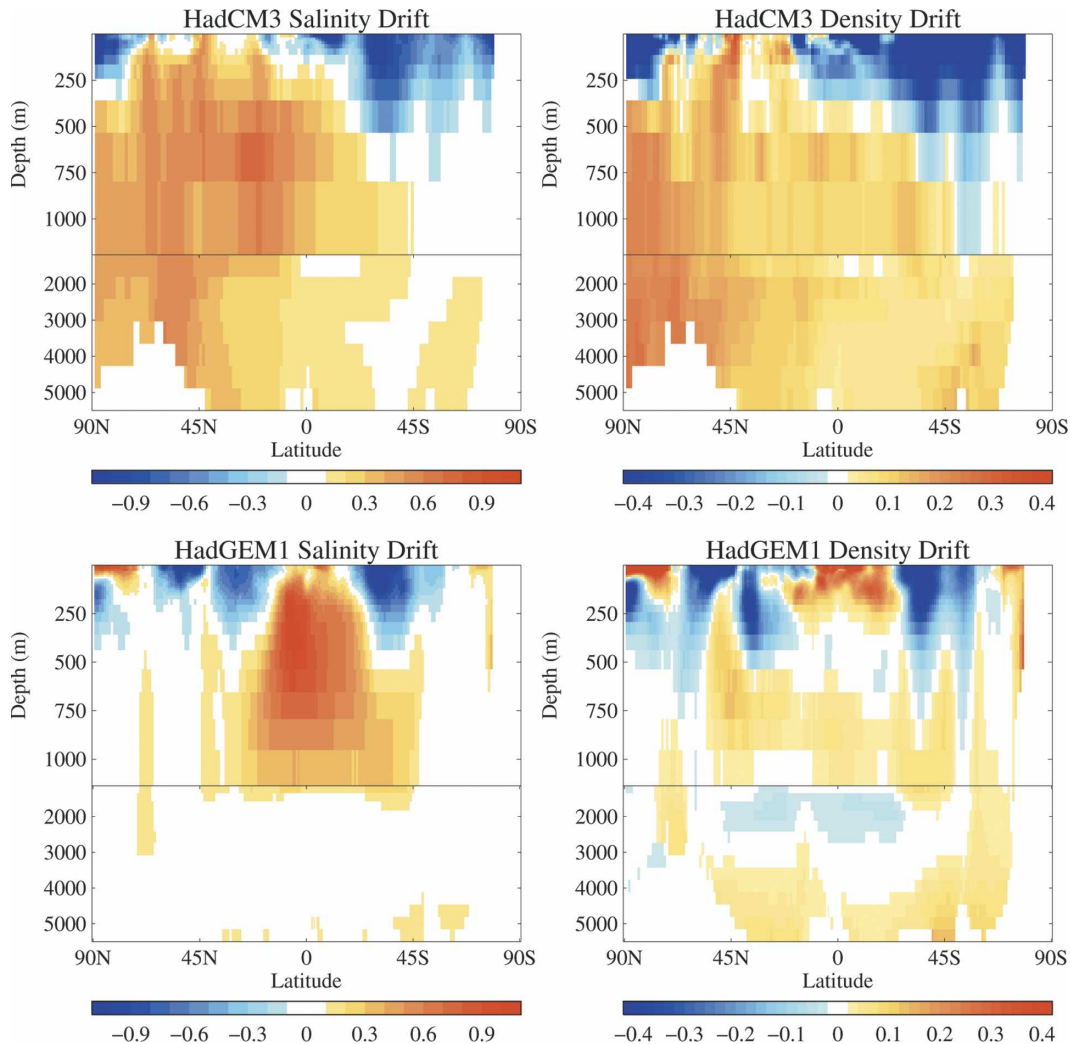


FIG. 8. Drifts in Atlantic/Arctic zonal mean salinity (psu) and density (kg m^{-3}) vs depth for (top) HadCM3 and (bottom) HadGEM1, relative to the first year of the simulations, for the same periods as in Fig. 2.

HadGEM1 are shown in Fig. 10 where they are compared with a range of observational estimates as described in Roberts et al. (2004). Overall the ocean heat transports in both models are within the error bars of observational estimates. Figure 10 also shows the components of the heat transport with the gyre and meridional (or overturning) components as described in Hall and Bryden (1982). The largest difference between the heat transport simulation of HadCM3 and HadGEM1 is that the Atlantic heat transport in HadCM3 is dominated by the meridional component (as suggested by Hall and Bryden 1982), while in HadGEM1 the meridional component of the heat transport is reduced between 20° and 40°N and the gyre component is increased.

The overturning streamfunction in potential temperature space (Fig. 11) shows how the net heat transport is accomplished; in HadCM3 warm water is trans-

ported northward and is almost all converted into NADW in agreement with the observations of Hall and Bryden (1982). In HadGEM1, the NADW cell continues to dominate but is supplemented by a shallow overturning cell between 20° and 40°N centered at 20°C . The shallow cell is not visible in Fig. 9 because it takes place as a horizontal circulation with warm northward flow in the western Atlantic and cold southward flow in the eastern basin. A weak shallow circulation cell is evident in HadCM3 but makes little contribution to the heat transport, while in HadGEM1 it makes a significant contribution because it converts water from 20° to 15°C .

e. Sea ice

The Northern Hemisphere wintertime pressure at mean sea level in the atmosphere submodel of

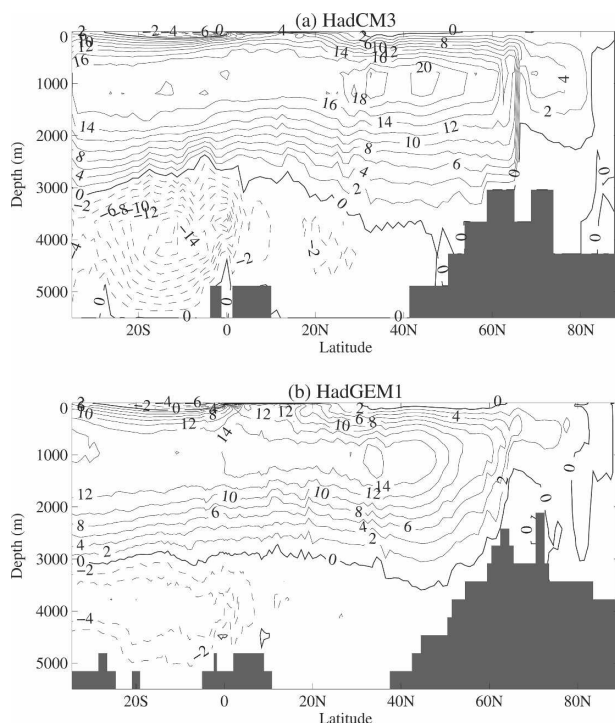


FIG. 9. Atlantic overturning streamfunction (S_v) for (a) HadCM3 and (b) HadGEM1, for the same periods as in Fig. 2.

HadGEM1 is more realistic than in HadCM3 (Martin et al. 2006), leading to an improvement in surface wind forcing of the sea ice submodel. A consequence of the superior wind forcing, together with the improved ice dynamics, is a more realistic Northern Hemisphere ice thickness distribution in HadGEM1, compared with HadCM3 (Fig. 12). In particular, the thickest ice in HadGEM1 correctly banks up against the northern coasts of Greenland and the Canadian Archipelago, whereas in HadCM3 the thickest ice is located in the Beaufort gyre.

The HadGEM1 ice extent generally compares well with observations, although the winter ice is too extensive in the northernmost Pacific, consistent with HadGEM1's cold SST bias in this region (Fig. 3). The pattern of ice motion also compares well with observations, although the ice speeds are generally too fast. For a more complete evaluation of the simulated sea ice in HadGEM1 see MCL.

7. Tropical variability

a. El Niño–Southern Oscillation (ENSO)

The simulation of ENSO is HadGEM1's main weakness as a coupled model. In HadGEM1 the climatological trade winds are too strong westward of about

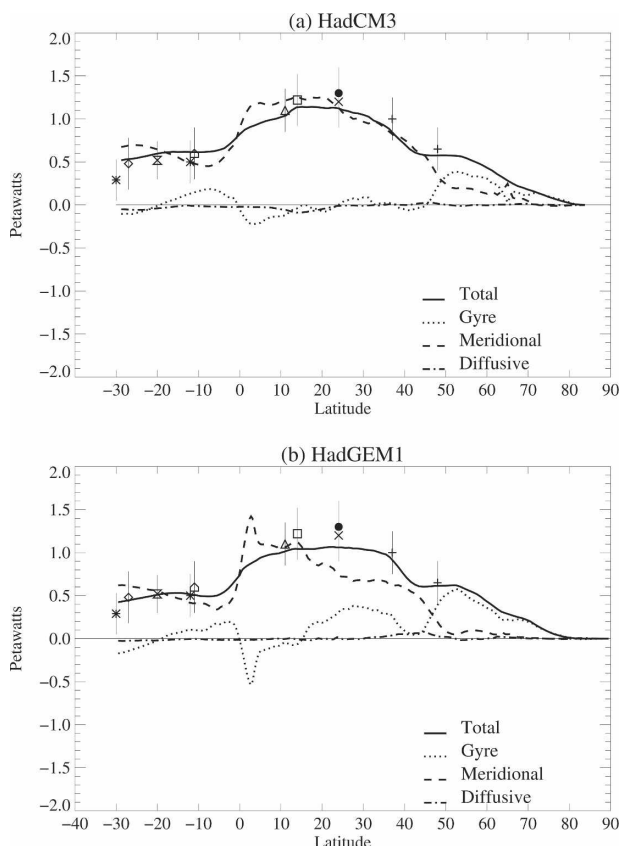


FIG. 10. Modeled Atlantic heat transport for (a) HadCM3 and (b) HadGEM1 for the same periods as in Fig. 2. Observational estimates of ocean heat transport and the associated error bars are also shown.

150°W, and the excessive zonal wind stress in the equatorial tropical Pacific (Fig. 13a) then drives excessive upwelling across much of the tropical Pacific. We believe this to be a major cause of the deficiencies with ENSO, although it may not be the only factor.

The effect of the excessively strong trade winds in HadGEM1 is to depress equatorial SSTs (Fig. 13b) and push the thermocline too deep in the western Pacific warm pool. In fact, the thermocline in HadGEM1 is about 20–25 m too deep in December in the western tropical Pacific compared with an ocean analysis (not shown). The thermocline depth in the western tropical Pacific in HadCM3 and HadCEM was very close to the ocean analysis, by contrast, reflecting the fact that the equatorial wind stress forcing was closer to reality in those models.

The standard deviation of variability of Niño-3 (5°S–5°N, 150°–90°W) SST monthly mean anomalies is somewhat low in HadGEM1 at 0.69 compared to 0.80 K in the Hadley Centre Sea Ice and SST (HadISST) dataset (1873–2002; Rayner et al. 2003). HadCM3

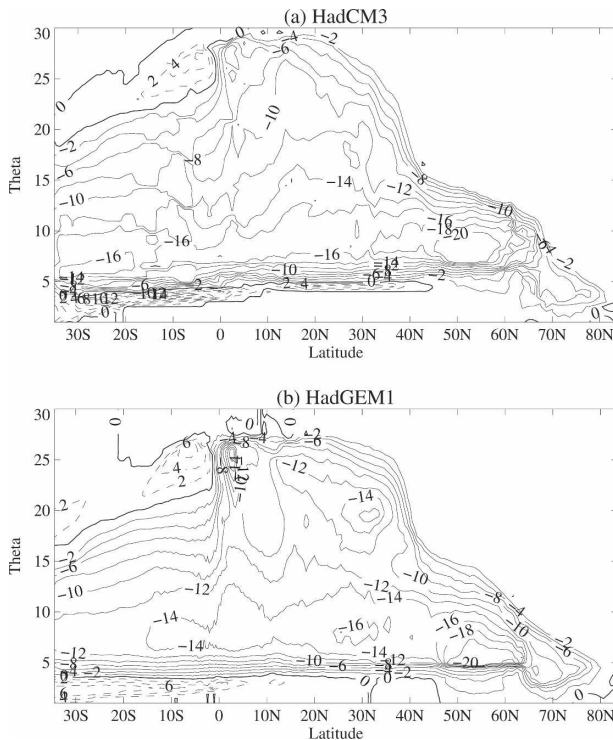


FIG. 11. As in Fig. 9 but for streamfunction (Sv) plotted in latitude vs potential temperature space.

(0.85 K) and HadCEM (1.12 K) both overestimated the variability by this measure.

Most observed ENSO events begin in northern spring and peak from November to January. The annual cycle of Niño-3 SST interannual variability pro-

vides a measure of this behavior. The HadGEM1 simulation does not capture the observed phase locking well (Fig. 14a). Nonetheless the HadGEM1 power spectrum (Fig. 14b) shows a dominant narrow peak at about 4 yr, which is close to the major time scale observed. However, the lack of power at other time scales suggests that its ENSO cycle may be too regular.

In general, the positive-phase SST anomalies in the central and eastern tropical Pacific are weaker in HadGEM1 than in observations, HadCM3, or HadCEM. National Centers for Environmental Prediction (NCEP) analyses of composite December–February (DJF) El Niño sea level pressure anomalies indicate high pressure anomalies over the Maritime Continent and low pressure over the east tropical Pacific. These pressure anomalies are reproduced more weakly in HadGEM1, consistent with its weaker SST anomalies (not shown).

The east–west dipole structure of observed precipitation anomalies indicates that the convection maximum shifts eastward during El Niño winters. This is not captured in HadGEM1. In fact, the HadGEM1 control simulation produces enhanced precipitation at the equator and south of the equator across the tropical Pacific (not shown). This is probably due to the combined effect of the SST cold bias and weak SST variability in the tropical Pacific since HadGAM1 (with prescribed SSTs) reproduces the observed features in a more satisfactory manner (see section 7b).

In terms of El Niño dynamics, previous work describes two types of modes that give rise to El Niño, that is, the SST mode and thermocline mode (e.g., Nee-

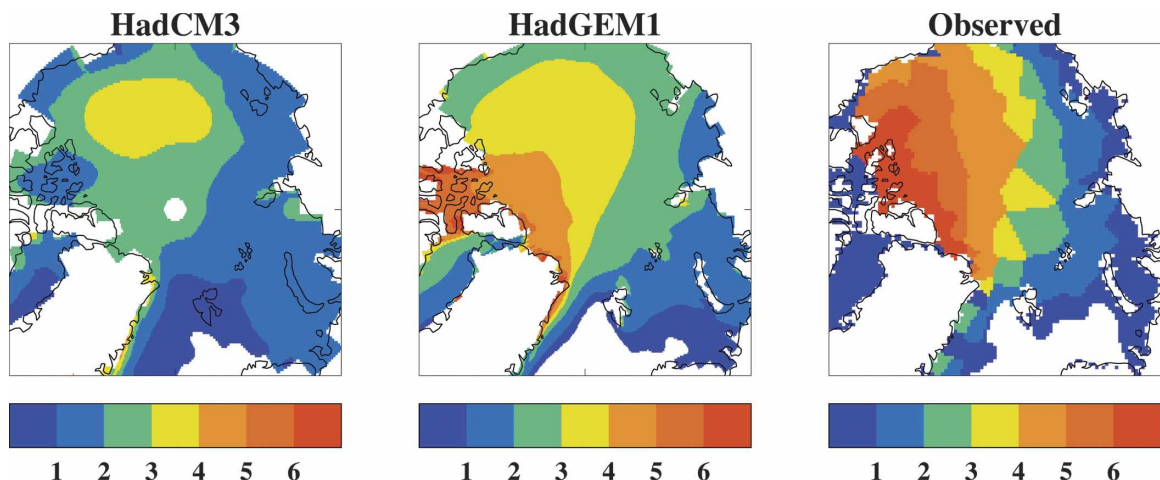


FIG. 12. Mean January to March ice thickness (m) over the Arctic from (left) HadCM3, (middle) HadGEM1, and (right) observations. The model data are for the same periods as in Fig. 2. The observations have been interpolated from submarine upward-looking sonar data (Bourke and Garrett 1987). Values are only shown where the sea ice concentration is greater than 0.15. Sea ice concentrations from HadISST (Rayner et al. 2003) for 1979–2002 are used to mask the submarine observations.

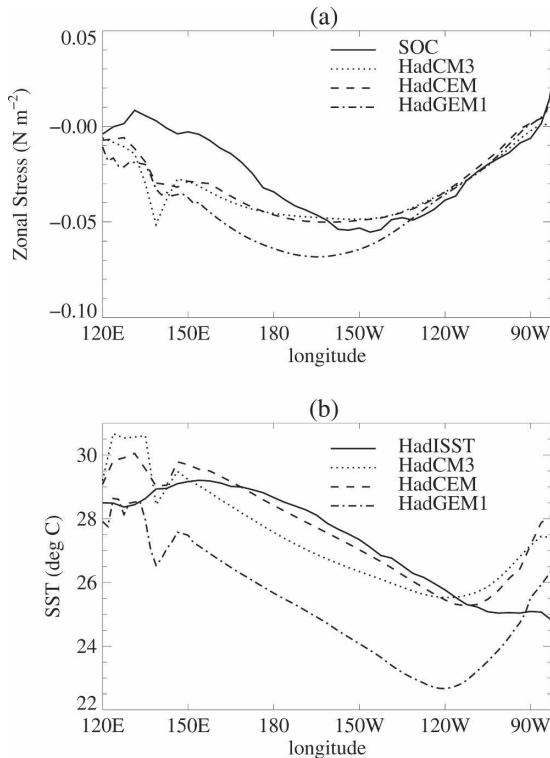


FIG. 13. (a) Annual mean zonal wind stress and (b) tropical Pacific SSTs in HadCM3, HadCEM, HadGEM1, and climatological data (SOC and HadISST, respectively). Comparisons are based on 130 years of data except for the climatological zonal wind stress, which is based on 50 years only.

lin et al. 1998). The SST mode results from local wind–SST interaction in the central and eastern tropical Pacific and shows surface east to west propagation of SST anomalies, whereas the thermocline mode results from remote wind–thermocline feedbacks involving the west tropical Pacific and shows west to east propagation of subsurface ocean temperature anomalies. The thermocline mode behavior bears some similarities to the “delayed oscillator” mechanism (Zebiak and Cane 1987; Suarez and Schopf 1988).

The weak subsurface temperature anomalies before the peak of El Niño in HadGEM1 (Fig. 15) indicate a weak remote impact from the west tropical Pacific, while time–longitude cross sections of composite SST anomalies in the equatorial 5°S – 5°N latitude band of the central Pacific (not shown) indicate westward propagation in HadGEM1. These facts suggest that the dynamics of El Niño in HadGEM1 are similar to the SST mode (SST anomalies being predominantly due to the local wind–SST interaction) rather than the thermocline mode or “delayed oscillator” mechanism. More work is required to understand why the remote

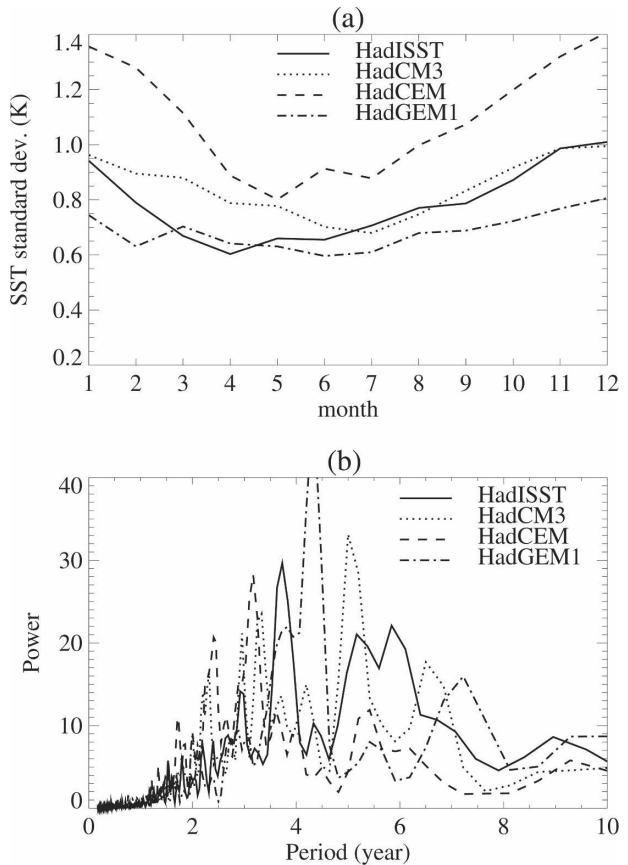


FIG. 14. (a) Annual cycle of the interannual standard deviation of monthly mean Niño-3 SSTs for HadISST, HadCM3, HadCEM, and HadGEM1; (b) power spectra of the corresponding Niño-3 monthly mean anomaly time series; 130 years of data are used in Fig. 13.

wind–thermocline feedback in HadGEM1 appears to be too weak.

b. Cloud–climate interactions during El Niño

The simulation of clouds in HadGEM1 is described by Martin et al. (2006). There has been considerable recent interest in devising ways of testing cloud–climate interactions within models using, for example, cloud changes in response to short-term forcing such as strong El Niño events. Cess et al. (2001) show that the lack of a zonal SST gradient in the tropical Pacific Ocean during the 1997–98 El Niño caused a collapse of the Walker circulation and enhanced upward motion over the tropical eastern Pacific (TEP). Using satellite data from the Earth Radiation Budget Experiment (ERBE) and the Clouds and the Earth’s Radiant Energy System (CERES), Cess et al. (2001) show that these circulation changes are associated, on average, with higher-level clouds in the TEP and lower clouds in the tropical west-

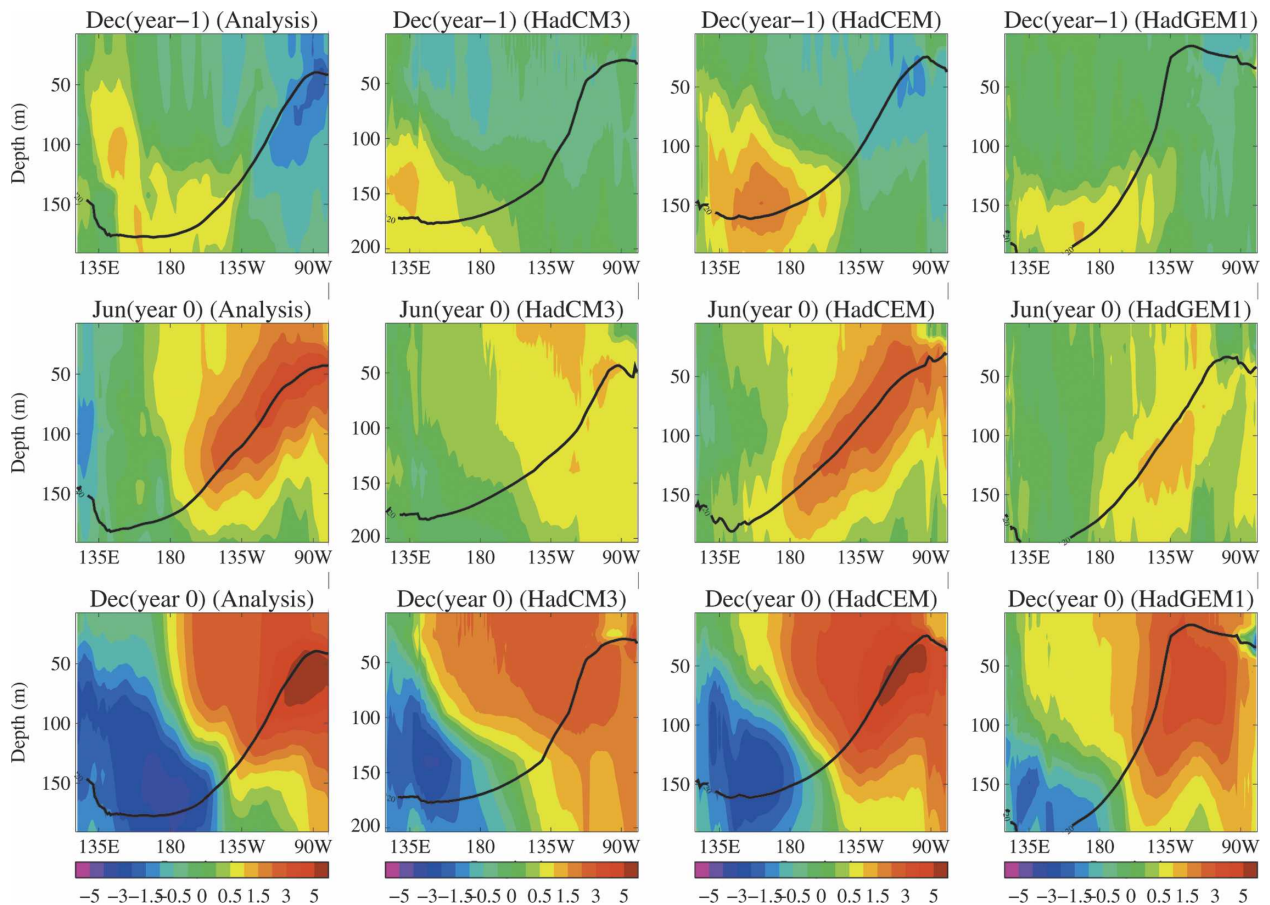


FIG. 15. Evolution of upper-ocean temperature anomalies ($^{\circ}\text{C}$) and 20°C isotherm (thick line) for composite El Niño events along the equatorial tropical Pacific (2.5°S – 2.5°N) in HadGEM1, HadCM3, and HadCEM and the analysis of actual subsurface temperatures using the Simple Ocean Data Assimilation (SODA) of Carton et al. (2000). The comparison is based on 50 years of data.

ern Pacific (TWP); that is, clouds move up in the TEP and down in the TWP. Lu et al. (2004) show that HadAM3 reproduced both the collapse of the Walker circulation and similar trends in cloud altitudes over both the TEP and TWP as seen in the satellite data. Here we show the same analysis for HadGAM1 when run in AMIP-II mode. This simulation also reproduces the collapse of the Walker circulation (Fig. 16) between 1985 (a “normal” year) and 1998 (a strong El Niño year) as seen in the NCEP reanalysis data (Fig. 4 of Cess et al. 2001). The cloud response can be characterized by the (sign-reversed) ratio of shortwave (SW) to longwave (LW) cloud forcing, N , here shown from ISCCP radiative fluxes (Zhang et al. 2004; Table 3). In a typical year, over the TWP there is a reasonable amount of compensation between SW and LW cloud forcing, but in 1998 SW cooling dominates and N increases (Table 3). In the TEP, SW cooling dominates in a typical year, but there is a strong reduction in 1998 with N being lower than in the TWP. HadGAM1 cap-

tures the shift in convection from west to east in the tropical Pacific and simulates the changes in N realistically (Table 3).

A preliminary assessment of the degree to which these same processes are captured in the coupled model HadGEM1 (by compositing strong El Niño events—here chosen to be events defined by a preceding December Niño-3 SST anomaly greater than three standard deviations of the monthly Niño-3 SST index) shows very little change in N in either basin (Table 3) and no evidence of a collapse of the Walker circulation in the zonal winds (not shown). This lack of response is likely to be associated with overly weak SST variability and the cold bias in the tropical Pacific. The combined effect is that SSTs are not warm enough to trigger convection in the central and eastern tropical Pacific, even during El Niños, and this is a subject of future study. In contrast, HadCM3 reproduces the collapse of the Walker circulation associated with El Niños (not shown). Changes of N in HadCM3 in the TWP and TEP

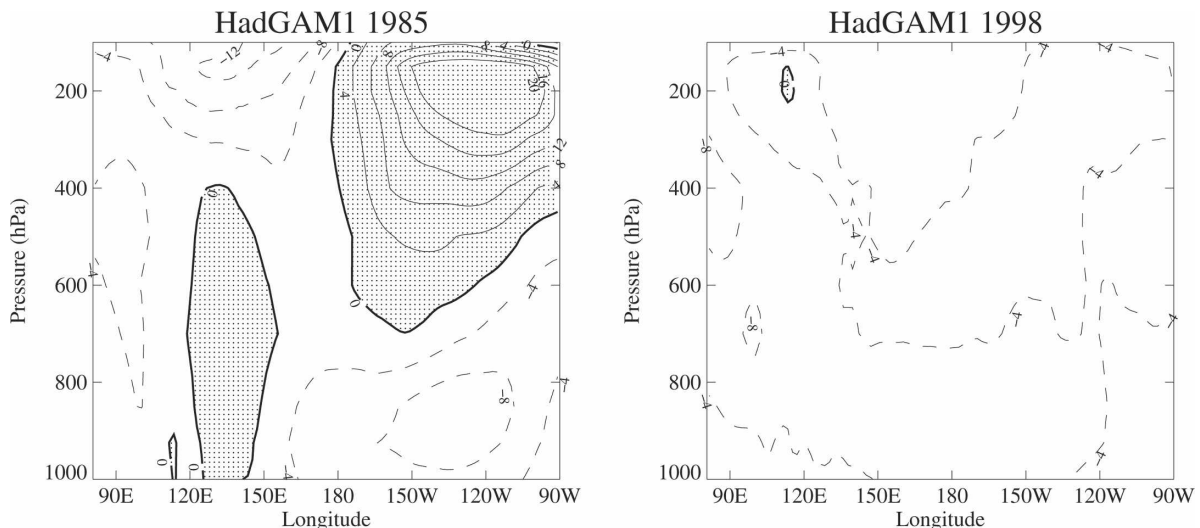


FIG. 16. Pressure–longitude cross sections of zonal mean wind (m s^{-1}) averaged from 5°S to 5°N from a HadGAM1 AMIP-II simulation for a 4-month (JFMA) mean for (left) 1985—a normal year—and (right) 1998—a strong ENSO year.

are also in better agreement with observations (Table 3) although the magnitude of these changes is smaller, presumably due to the westward extension of positive SST anomalies associated with El Niños in HadCM3 (not shown).

c. Indian Ocean variability

A basic analysis of the interannual variability of the tropical Indian Ocean with particular reference to the Indian Ocean dipole (IOD) or zonal mode (Saji et al. 1999; Webster et al. 1999) has been completed using 52 years from the HadGEM1 spinup phase of the control run, in comparison with HadCM3 and observed SSTs from HadISST. This analysis builds on the comprehensive diagnosis of three coupled models—HadCM3,

HadCEM and a low horizontal resolution ocean version of HadCEM (HadCEML)—described in Spencer et al. (2005). In this paper only SST is analyzed, subsurface fields being excluded.

El Niño is the dominant factor in the observed seasonal mean interannual variability in tropical SST and also to a large extent in HadCM3 (Fig. 17). In HadGEM1, however, El Niño variability appears to be somewhat reduced and the seasonality, with maximum variance in MAM, is clearly wrong. As in HadCM3, the SST variability in the subtropics of the North Pacific and southern Indian Ocean is greatly overestimated for reasons that are unclear at present, although they are not thought to be ENSO related. In the equatorial Indian Ocean, HadGEM1 has a high level of variability along the coasts of Java and Sumatra during June–August (JJA) and September–November (SON), indicative of an overactive IOD.

The evolution of observed IOD events and those in HadGEM1 and HadCM3 are shown in Fig. 18 as the dipole index (the difference in SST between the west and southeast tropical Indian Ocean) for two years centered on each October when the index is greater than one standard deviation above the long-term mean for each series. The composites are based on the October index because this is the month when the IOD generally peaks. Both models have realistic timing of the onset and demise of these events. IOD events are not observed to occur strongly for two years in a row, but this was a very common occurrence in HadCM3 [attributable to the low ocean resolution according to Spencer et al. (2005)]. This problem is largely eliminated in HadGEM1, which has higher ocean resolution. Only two

TABLE 3. The value of N (see text) for ISCCP-FD satellite observations (Zhang et al. 2004) for 1985 (normal year) and 1998 (strong ENSO), HadGAM1 for 1985 and 1998, HadCM3 and HadGEM1 for a 130-yr climatology (normal year), and, respectively, one and a composite of two strong ENSO events (defined as events greater than three standard deviations away from climatology). The regions TWP and TEP are as used by Cess et al. (2001). The percentage change from normal year is shown in brackets.

Model	TWP		TEP	
	Normal year	Strong ENSO	Normal year	Strong ENSO
ISCCP	1.27	1.84 (+45%)	2.7	1.63 (−40%)
HadCM3	1.48	1.67 (+13%)	2.7	1.95 (−28%)
HadGAM1	1.20	1.47 (+23%)	2.7	1.44 (−47%)
HadGEM1	1.29	1.25 (−3%)	3.0	2.62 (−11%)

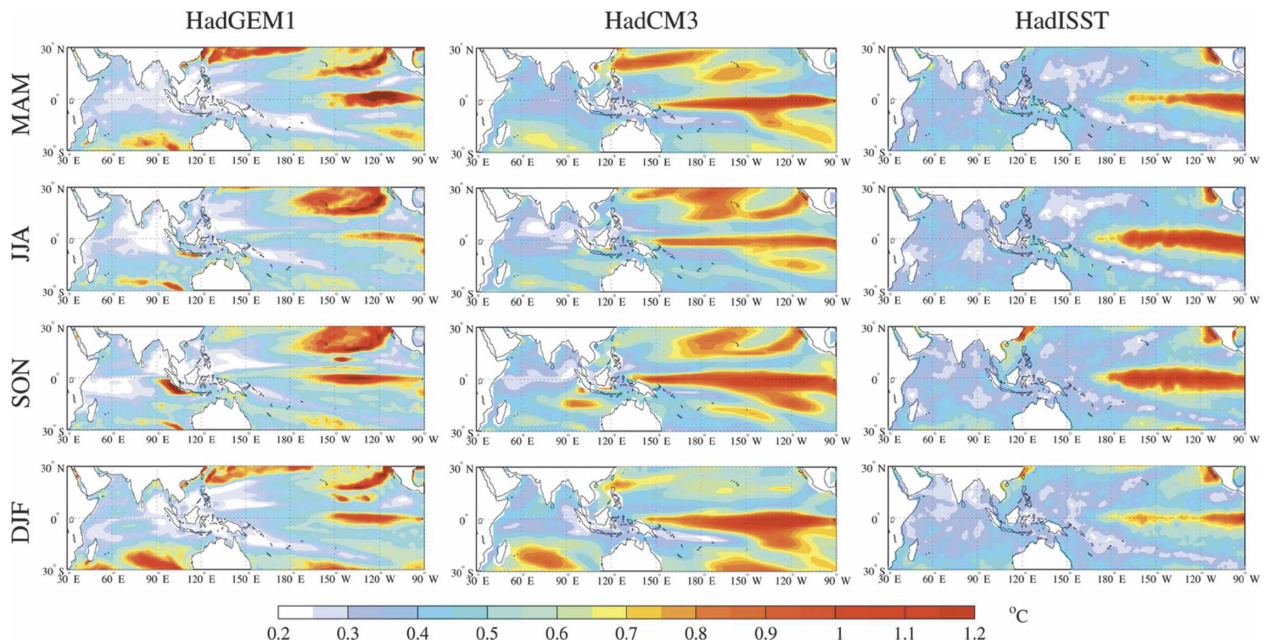


FIG. 17. Standard deviation of the seasonal mean interannual variability in SST (K) from a 52-yr section of the HadGEM1 control run and a 100-yr section of the HadCM3 control run, compared with HadISST (Rayner et al. 2003; not detrended) for 1873–2004.

of the ten events reoccur the following year in HadGEM1 with a composite value in the following October close to zero (similar statistics to HadISST), whereas more than half of the events reoccur the following year in HadCM3 with a composite value the following October of nearly 0.5 K. However, the amplitude of the seasonal variation in IOD is even more marked in HadGEM1 than HadCM3 (see also Fig. 17), suggesting an overly strong positive feedback between the southeasterly winds along the coast of Sumatra, coastal upwelling, and cooler SSTs during SON. This is a common problem among models with similar ocean resolution [e.g., the upgraded version of the Scale Interaction Experiment (SINTEX-F1) model; Yamagata et al. 2004].

The IOD has a strong association with El Niño in observations (Fig. 19). The atmospheric teleconnections into the Indian Ocean region give rise to a basin-wide warming, which is evident in the large, positive correlations between Niño-3.4 and the western node of the IOD in the observations. In contrast, the correlation in HadGEM1 between Niño-3.4 and the western node of the IOD is very weak for most of the year, particularly in northern spring, suggesting that the remote effects of ENSO are not as well captured as in HadCM3. It is worth noting that the lack of correlation between Niño-3.4 and the Indian Ocean during the Asian summer monsoon season in HadCM3 can be substantially improved when the systematic SST errors in the western and central Pacific are reduced (Turner et al. 2005).

Both models capture the strong negative correlation in northern autumn between El Niño and the SSTs in the southeast Indian Ocean—the southeast node of the IOD. Overall, the relationship between ENSO and the IOD is represented to some extent in both models with the strongest correlations occurring toward the end of the year. However the abrupt decline in correlation is not well captured, suggesting that the termination of the IOD by the seasonal reversal of the monsoon winds is not well simulated in either model.

Further research is planned to explore the vertical structure of the Indian Ocean in the model and the characteristics of the ocean waves that contribute to the IOD and to Indian Ocean variability in general. The earlier study of Spencer et al. (2005) suggested that enhanced vertical resolution in the ocean and the proper representation of Java and Sumatra are key factors in capturing the IOD. Both of these elements are present in HadGEM1.

8. Climate sensitivity

Despite numerous differences in model formulation between HadCM3 and HadGEM1, the transient and effective climate sensitivities, T_t and T_e respectively (Cubasch et al. 2001), in the two models are similar, as are global mean forcings due to a doubling of CO₂ concentration (Table 4). Following Boer and Yu (2003), a local feedback analysis is presented that permits geographical maps of the feedbacks to be exam-

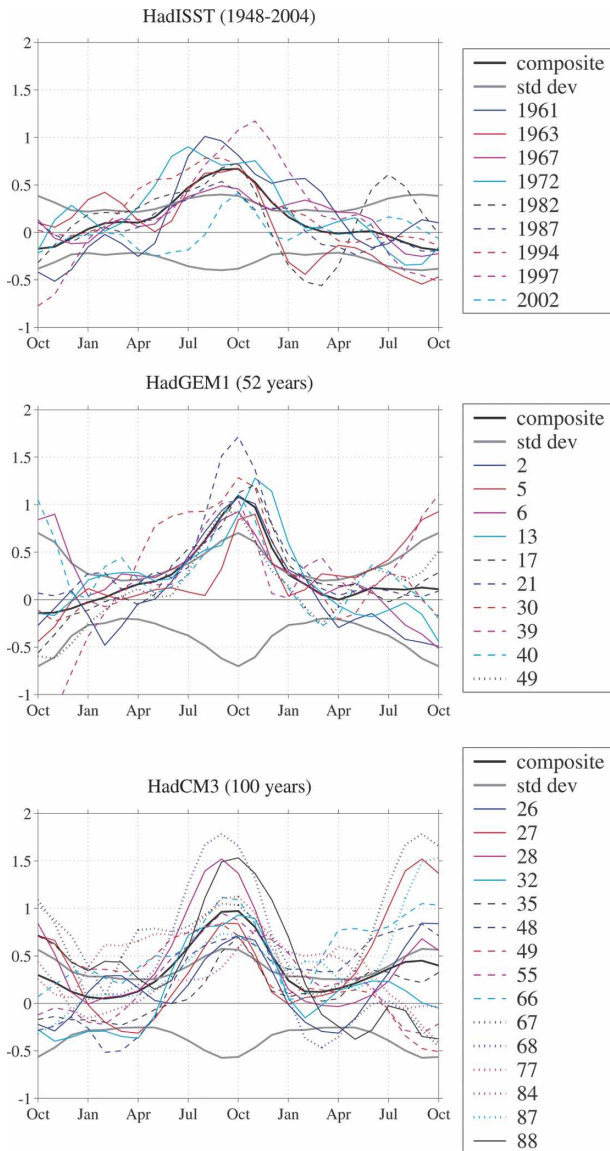


FIG. 18. The IOD index (events shown are for the years identified in which the index was greater than one standard deviation in October plus composites of such events) for HadISST (linearly detrended for 1948–2004), HadGEM1 (52-yr section of control run), and HadCM3 (100-yr section of control run). The index is based on the difference in SST between the west (10°S – 10°N , 50° – 70°E) and southeast (5°S – 0° , 90° – 110°E) tropical Indian Ocean and has been smoothed with a 1–2–1 binomial filter by month to suppress intraseasonal variability, which is large in the Indian Ocean. The standard deviation for each month of the year is also shown.

ined in addition to the global mean analysis. The method defines a local feedback parameter (Λ) as the local radiative imbalance at the top of the atmosphere (R' ; at equilibrium, the global mean of R' is, on average, zero) minus the local radiative forcing (f), which we evaluate using a double call to radiation routines,

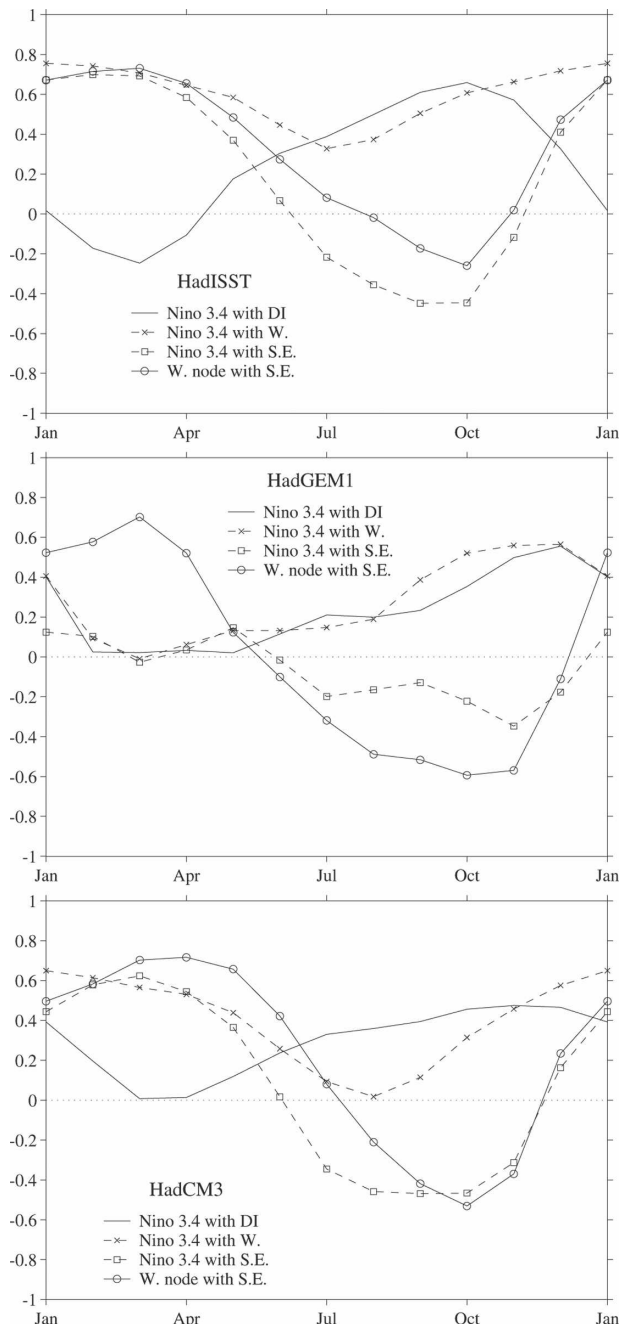


FIG. 19. Correlations in (top) HadISST, (middle) HadGEM1, and (bottom) HadCM3 for each month of the year between the Niño-3.4 index and the IOD index, between the Niño-3.4 index and the west and southeast nodes of the dipole index separately, and between the west and southeast nodes of the dipole. All datasets have been smoothed to suppress intraseasonal variability as in Fig. 18. The observational data (HadISST, from 1948 to 2003) have also been linearly detrended.

divided by the global mean change in surface temperature ($\langle T' \rangle$). The use of global rather than local temperature change in the denominator means that this can be thought of as the local contribution to the global feed-

TABLE 4. Global mean forcing ($\langle f \rangle$, W m^{-2}), transient climate sensitivity ($\langle T'_t \rangle$, K), effective climate sensitivity ($\langle T'_e \rangle$, K), and decomposed feedback parameters ($\langle \Lambda \rangle$, $\text{W m}^{-2} \text{K}^{-1}$) for HadGEM1 and HadCM3. These were calculated from 20-yr means centered on the time of CO_2 doubling in transient simulations with CO_2 increased at 1% per annum (CMIP experiments).

Model	$\langle f \rangle$	$\langle T'_t \rangle$	$\langle T'_e \rangle$	$\langle \Lambda \rangle$	$\langle \Lambda_s \rangle$	$\langle \Lambda_L \rangle$	$\langle \Lambda_A \rangle$	$\langle \Lambda_C \rangle$	$\langle \Lambda_{SA} \rangle$	$\langle \Lambda_{SC} \rangle$	$\langle \Lambda_{LA} \rangle$	$\langle \Lambda_{LC} \rangle$
HadCM3	3.86	2.0	3.1	-1.28	0.84	-2.12	-1.50	0.22	0.67	0.17	-2.16	0.04
HadGEM1	3.83	1.8	2.8	-1.39	0.86	-2.26	-1.54	0.15	0.76	0.10	-2.30	0.05

back parameter. The same notation is used as in Boer and Yu (2003) so that subscripts C and A denote cloud and cloud-free atmosphere feedbacks, respectively, and subscripts s and L denote shortwave and longwave feedbacks. Hence,

$$\Lambda = \Lambda_A + \Lambda_C = \Lambda_s + \Lambda_L = \Lambda_{SA} + \Lambda_{SC} + \Lambda_{LA} + \Lambda_{LC}.$$

In addition to the global mean being similar in the two models (accounting for the similarity in T_e), the global mean cloud and clear-sky, shortwave and longwave components of Λ are also very similar (Table 4). However, this global mean analysis disguises differences in the geographical pattern of the global-mean

surface temperature response (Fig. 20a). While the geographical patterns of the forcing are reasonably similar (not shown), the local feedbacks are somewhat different, particularly in the Tropics (Fig. 20b). There is a weaker negative feedback in HadGEM1 (positive difference between the models) over the southern tropical trade cumulus regions leading to greater warming here, and weaker positive feedbacks over the Pacific stratocumulus region and over parts of the Amazon region, resulting in reduced warming over these areas. These differences in the geographical pattern of the feedbacks can be seen to result mainly from differences in the pattern of cloud feedbacks (Fig. 20c) and largely from

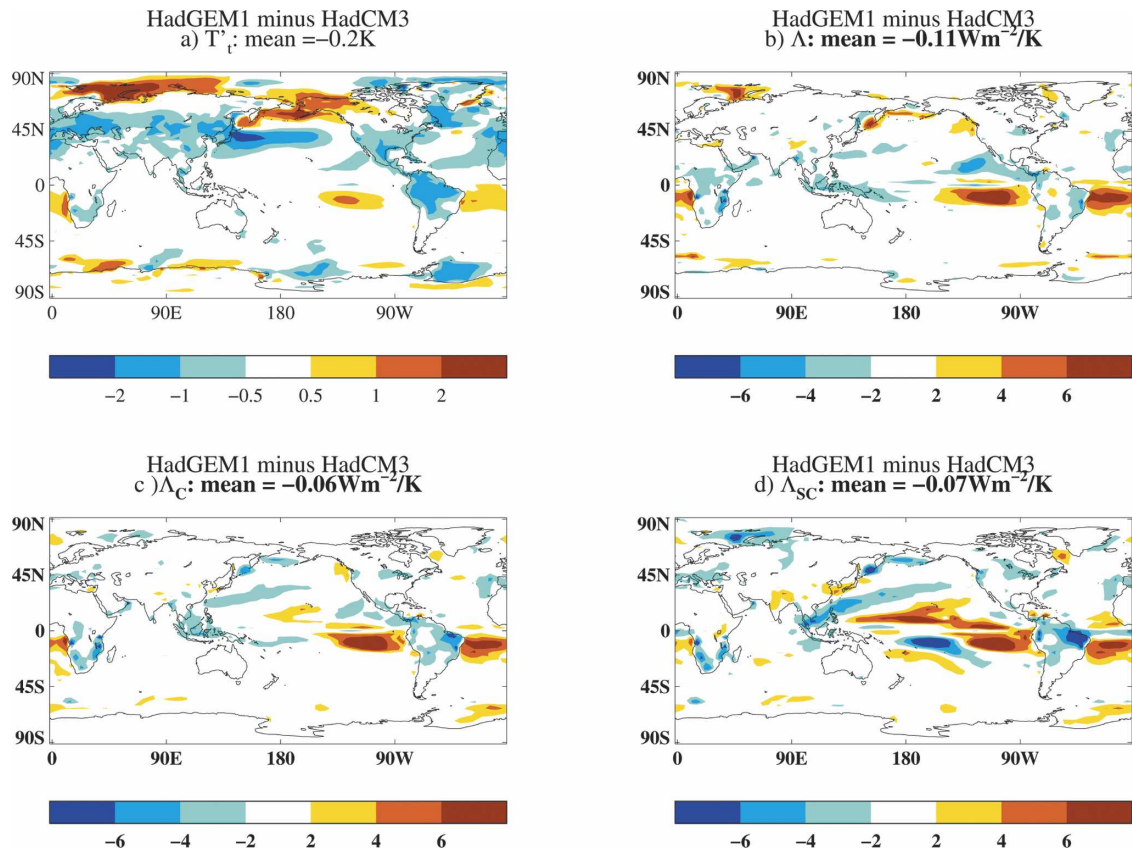


FIG. 20. The differences between HadGEM1 and HadCM3 for a 20-yr period centered at the time of CO_2 doubling from their respective CMIP experiments in (a) surface temperature response relative to the corresponding control run, (b) feedback parameter (Λ), (c) cloud feedback parameter (Λ_C), and (d) shortwave cloud feedback parameter (Λ_{SC}).

TABLE 5. The effect of changes in model formulation on the equilibrium global mean climate sensitivity ($\langle T'_{2xCO_2} \rangle$, K), and decomposed feedback parameters ($\langle \Lambda \rangle$, $\text{W m}^{-2} \text{ K}^{-1}$). Note that changes in $\langle \Lambda \rangle$ of less than $0.1 \text{ W m}^{-2} \text{ K}^{-1}$ are within the range of model internal variability.

Effect of changes (listed below) on:	$\langle T'_{2xCO_2} \rangle$	$\langle \Lambda \rangle$	$\langle \Lambda_S \rangle$	$\langle \Lambda_L \rangle$	$\langle \Lambda_A \rangle$	$\langle \Lambda_C \rangle$	$\langle \Lambda_{SA} \rangle$	$\langle \Lambda_{SC} \rangle$	$\langle \Lambda_{LA} \rangle$	$\langle \Lambda_{LC} \rangle$
New dynamical core	+0.2	+0.06	+0.30	-0.24	-0.05	+0.12	+0.07	+0.23	-0.12	-0.11
Increased horizontal resolution	-0.3	-0.04	+0.04	-0.07	-0.09	+0.05	-0.04	+0.08	-0.05	-0.02
Increased vertical resolution	-0.5	-0.16	-0.07	-0.09	+0.04	-0.20	+0.03	-0.10	+0.01	-0.10
New boundary layer	-0.1	+0.10	+0.06	+0.04	+0.11	-0.01	+0.04	+0.03	+0.07	-0.04
MOSES-II	-0.2	-0.03	+0.01	-0.04	-0.07	+0.03	-0.05	+0.06	-0.01	-0.03
Shallow convection	-0.3	-0.04	-0.01	-0.04	-0.02	-0.02	+0.04	-0.05	-0.06	+0.02
Convective anvils	+0.1	+0.03	+0.20	-0.18	-0.09	+0.12	-0.04	+0.24	-0.05	-0.13
Variable RH_{crit}	+0.2	+0.10	+0.02	+0.08	+0.02	+0.07	-0.02	+0.04	+0.04	+0.04
Vertical grad cloud area*	-0.6	-0.18	-0.07	-0.11	-0.07	-0.11	+0.02	-0.08	-0.08	-0.03
Cloud microphysics*	+0.3	+0.09	+0.11	-0.02	-0.00	+0.10	-0.02	+0.13	+0.01	-0.03
Aerosols	-0.4	-0.07	-0.06	-0.01	+0.00	-0.07	+0.03	-0.09	-0.03	+0.02
Nonspherical ice**	+0.4	+0.03	-0.22	+0.24	+0.15	-0.12	+0.01	-0.23	+0.13	+0.11
EVP sea ice dynamics*	-0.2	-0.03	-0.01	-0.02	+0.00	-0.03	+0.02	-0.02	-0.02	-0.01
ITD thermodynamics	+1.0	+0.25	+0.16	+0.08	+0.30	-0.05	+0.28	-0.12	+0.02	+0.07
Residual: minor changes (plus nonlinearities)**	+1.4	+0.24	-0.28	+0.50	+0.25	-0.01	+0.01	-0.28	+0.23	+0.27
HadGSM1 – HadSM3	+1.1	-0.33	+0.21	+0.12	+0.44	-0.11	+0.38	-0.17	+0.06	+0.06

* The effect of the parameterization change was derived assuming linearity with one other change.

** The effect was calculated as a residual after accounting for the other major change (i.e., including the possible effects of nonlinearity).

differences in the shortwave cloud feedback (Fig. 20d). There is little difference in the longwave cloud feedback over these regions (not shown), suggesting that a differing response of low cloud accounts for much of the different pattern of the feedbacks between the two models (consistent with the findings of Webb et al. 2006). Over the tropical central and western Pacific, the changes in shortwave and longwave cloud feedbacks largely cancel, suggesting a different response of deep convective cloud between the two models here, with little net radiative impact.

A series of sensitivity experiments have been performed using atmosphere models coupled to mixed layer ocean (slab) models to test the effect of each of the principal structural changes on the climate sensitivity. For coding reasons, the tests of each change used variously HadSM3, HadGSM1 (the slab model versions of HadCM3 and HadGEM1), or an intermediate slab model, HadSM4 (Williams et al. 2003), as their control. The experiments took one of these forms: HadSM3 plus a structural change (results being compared with HadSM3); HadSM4 minus a structural change (compared with HadSM4); HadGSM1 minus a structural change (compared with HadGSM1). Here, the “effect” of particular changes is presented, but it should be noted that a different effect may have resulted from an experiment testing HadSM3 plus a particular change compared with HadGSM1 minus the change, due to

interactions with other schemes changed between the model versions.

Except for the ITD (subgrid-scale ice thickness distribution) scheme (discussed below) none of the changes tested individually appear to have a large impact on climate sensitivity, with few having an effect outside that expected from natural variability (Table 5). In addition, although there is a small contribution from some of the parameterization changes, none of the changes alone or in linear combination can account for the different pattern of the feedbacks shown in Fig. 20. This suggests that the differences in the pattern of feedbacks is either due to the numerous small changes applied during tuning of HadGEM1, through the feedbacks combining nonlinearly, or, most likely, the result of interactions between two or more changes, meaning that equal and opposite results are not obtained from including a change in one base model compared with removing it from another. For example, the inclusion of the new boundary layer parameterization (Lock et al. 2000) is found to have little impact on the feedbacks when introduced into HadSM3. Tests have indicated that the interaction of the new boundary layer scheme with ND account for much of the increase in stratocumulus in the present-day climate (Martin et al. 2006). Hence, a larger impact on the climate change response might be observed if the scheme were to be removed from HadGSM1, which is constructed around the ND

core (but this experiment has not been performed for coding reasons). When this boundary layer scheme was included in the Geophysical Fluid Dynamics Laboratory (GFDL) model, it was found to have a considerable impact on the feedback strength of tropical low cloud (B. Soden 2004, personal communication).

Table 5 shows a substantial impact on Λ_{SA} from the introduction of multiple sea ice thickness categories (ITD thermodynamics; MCL). The results in Table 5 are calculated from slab models, the increase in Λ_{SA} being much smaller in the coupled models. Consequently, the difference in the slab model equilibrium climate sensitivities is larger than the difference in the effective climate sensitivity of the coupled models ($T'_{2\times CO_2}$ is 3.3 K for HadSM3 and 4.4 K for HadGSM1). The ITD sea ice scheme in HadGEM1 is considerably more complex than in HadCM3 and is more sensitive to ocean dynamics and processes that are not represented in the slab model. As a result, HadGSM1 simulates sea ice that is too extensive in the control compared with HadGEM1. In contrast, there is less sea ice in the HadSM3 control than in HadCM3. While the sea ice scheme included in a slab model should be as close as possible to the parallel coupled model, an important consideration is to ensure that future schemes implemented in slab models also provide a similar sea ice distribution in the control and simulate similar feedback strength to their parallel coupled model.

Consistent with the transient climate sensitivities being similar in the two coupled models, changes in global mean precipitation at the time of CO_2 doubling are also similar (an increase of 2.0% in HadGEM1, compared with 1.8% in HadCM3) (Allen and Ingram 2002). The precipitation response at equilibrium in HadGSM1 is larger than in HadSM3 due to the greater equilibrium warming (7.9% in HadGSM1 compared with 5.8% in HadSM3).

9. Concluding discussion

We have evaluated the performance of a new Hadley Centre climate model, HadGEM1, against its predecessor HadCM3 and observations of recent climate. The new model continues the Unified Model strategy for climate and NWP at the Met Office within the framework of a semi-Lagrangian, nonhydrostatic dynamical core. The paper by Martin et al. (2006) demonstrates how the atmospheric performance of HadGEM1 improves on HadCM3 in important respects, benefiting from ND and major upgrades in its physical parameterizations and specifically from improvements in the representation of clouds and their effect on the radiation budget through the depth of the atmosphere. HadGEM1

also contains a much more advanced representation of sea ice dynamics and physics that, in concert with the improved forcing afforded by the atmosphere, leads to a better simulation in several ways.

We have described some problems apparent in HadGEM1 in its representation of the Tropics and simulation of ENSO, which we plan to investigate further. This work will focus for instance on the sensitivity of the errors in surface forcing (particularly wind stress) to changes in the atmospheric convection scheme, together with the ocean forced response and associated coupled feedbacks inherent in the model. On the other hand, HadGEM1 is able to capture at least some aspects of tropical variability better than HadCM3. For example, the Madden–Julian oscillation (MJO), which is the dominant mode of intraseasonal variability, is better represented in HadGEM1 than in HadCM3, as shown by Ringer et al. (2006). Neither model captures the propagating nature of MJO when run with prescribed SSTs, supporting the idea that the MJO is a fundamentally coupled atmosphere–ocean mode of variability (Woolnough et al. 2000).

Taken overall, therefore, HadGEM1 clearly performs well and in a quasi-objective sense (measured by the multielement CPI index we use) outperforms HadCM3 in simulating the present-day mean climate. Murphy et al.'s (2004) study suggests that models scoring better on the CPI turn out as better climate change predictors within a “perfect model” test framework. This, together with our belief that the theoretical basis of the parameterizations used in HadGEM1 is more physically realistic, gives us some confidence in HadGEM1's potential as a better climate predictor. HadGEM1 exhibits a similar climate sensitivity to CO_2 doubling as HadCM3. There are, however, significant differences in the geographical pattern of the response that are partly attributable to the better representation of marine stratocumulus in HadGEM1.

HadGEM1 already forms the basis for the development of a higher-resolution coupled model (HiGEM: Norton 2004) in a project supported by the U.K. Natural Environment Research Council. This project aims to deliver a community model to promote the understanding of climate processes by a wide user community and to facilitate reduction of systematic model errors. HadGEM1 is also being extended to encompass coupled climate, chemistry, and ecosystem modeling to permit Earth System experiments with a more consistent representation of feedbacks between climate, the biosphere, and chemistry than previously possible (although technical challenges and uncertainties in modeling these are considerable). Via collaborations with Japanese scientists (closely related to the HiGEM proj-

ect) we also plan to use the Earth Simulator supercomputer (Habata et al. 2003; Sato 2004) to explore Earth System experimental frontiers with HadGEM1 at high resolution over coming years.

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REFERENCES

- Acreman, D., 2005: The impact of improved vertical resolution and model tuning on mixed layer models. NCOF Tech. Rep. 2, National Centre for Ocean Forecasting, Met Office, Exeter, United Kingdom, 28 pp. [Available online at www.metoffice.gov.uk/research/ncof/pub/technical.html.]
- Allen, M. R., and W. J. Ingram, 2002: Constraints on future changes in climate and the hydrologic cycle. *Nature*, **419**, 224–232.
- Banks, H. T., 2000: Indonesian throughflow in a coupled climate model and the sensitivity of the heat budget and deep overturning. *J. Geophys. Res.*, **105**, 26 135–26 150.
- Bitz, C. M., M. M. Holland, A. J. Weaver, and M. Eby, 2001: Simulating the ice-thickness distribution in a coupled climate model. *J. Geophys. Res.*, **106**, 2441–2463.
- Boer, G. J., and B. Yu, 2003: Climate sensitivity and response. *Climate Dyn.*, **20**, 415–429.
- Bourke, R. H., and R. P. Garrett, 1987: Sea ice thickness distribution in the Arctic Ocean. *Cold Regions Sci. Technol.*, **13**, 259–280.
- Bryan, K., 1969: A numerical method for the study of the circulation of the world ocean. *J. Comput. Phys.*, **4**, 347–376.
- Carton, J. A., G. Chepurin, and X. Cao, 2000: A simple ocean data assimilation analysis of the global upper ocean 1950–95. Part II: Results. *J. Phys. Oceanogr.*, **30**, 311–326.
- Cess, R. D., M. H. Zhang, P.-H. Wang, and B. A. Wielicki, 2001: Cloud structure anomalies over the tropical Pacific during the 1997/98 El Niño. *Geophys. Res. Lett.*, **28**, 4547–4550.
- Cox, M. D., 1984: A primitive equation, three dimensional model of the ocean. Ocean Group Tech. Rep. 1, GFDL, Princeton, NJ, 143 pp.
- Cox, P. M., R. A. Betts, C. B. Bunton, R. L. H. Essery, P. R. Rowntree, and J. Smith, 1999: The impact of new land surface physics on the GCM simulation of climate and climate sensitivity. *Climate Dyn.*, **15**, 183–203.
- Cubasch, U., and Coauthors, 2001: Projections of future climate change. *Climate Change 2001: The Scientific Basis*, J. T. Houghton et al., Eds., Cambridge University Press, 525–582.
- da Silva, A. M., C. C. Young, and S. Levitus, 1994: *Algorithms and Procedures*. Vol. 1, *Atlas of Surface Marine Data*, NOAA Atlas NESDIS 6, 83 pp.
- Davies, T., M. J. P. Cullen, A. J. Malcolm, M. H. Mawson, A. Staniforth, A. A. White, and N. Wood, 2005: A new dynamical core for the Met Office's global and regional modelling of the atmosphere. *Quart. J. Roy. Meteor. Soc.*, **131**, 1759–1782.
- Davison, P. S., D. L. Roberts, R. T. Arnold, and R. N. Colville, 2004: Estimating the direct radiative forcing due to haze from the 1997 forest fires in Indonesia. *J. Geophys. Res.*, **109**, D10207, doi:10.1029/2003JD004264.
- Dickson, R. R., and J. Brown, 1994: The production of North Atlantic Deep Water: Sources, rates and pathways. *J. Geophys. Res.*, **99**, 12 319–12 341.
- Dukowicz, J. K., and R. D. Smith, 1994: Implicit free surface method for the Bryan–Cox–Semtner ocean model. *J. Geophys. Res.*, **99**, 7991–8014.
- England, M. H., 1995: The age of water and ventilation timescales in a global ocean model. *J. Phys. Oceanogr.*, **25**, 2756–2777.
- Essery, R. L. H., M. J. Best, R. A. Betts, P. M. Cox, and C. M. Taylor, 2003: Explicit representation of subgrid heterogeneity in a GCM land surface scheme. *J. Hydrometeorol.*, **4**, 530–543.
- ETOPO5, 1988: Global 5' × 5' depth and elevation. Tech. Rep. [Available from National Geophysical Data Center, NOAA, U.S. Dept. of Commerce, Code E/GC3, Boulder, CO 80303.]
- Gent, P. R., and J. C. McWilliams, 1990: Isopycnal mixing in ocean circulation models. *J. Phys. Oceanogr.*, **20**, 150–155.
- GFDL Global Atmospheric Model Development Team, 2004: The new GFDL Global Atmosphere and Land Model AM2–LM2: Evaluation with prescribed SST simulations. *J. Climate*, **17**, 4641–4673.
- Gordon, A. L., and R. Fine, 1996: Pathways of water between the Pacific and Indian oceans in the Indonesian seas. *Nature*, **379**, 146–149.
- Gordon, C., C. Cooper, C. A. Senior, H. T. Banks, J. M. Gregory, T. C. Johns, J. F. B. Mitchell, and R. A. Wood, 2000: The simulation of SST, sea ice extents and ocean heat transports in a version of the Hadley Centre coupled model without flux adjustments. *Climate Dyn.*, **16**, 147–168.
- Griffies, S. M., 1998: The Gent–McWilliams skew flux. *J. Phys. Oceanogr.*, **28**, 831–841.
- , A. Gnanadesikan, R. C. Pacanowski, V. D. Larichev, J. K. Dukowicz, and R. D. Smith, 1998: Isoneutral diffusion in a z-coordinate ocean model. *J. Phys. Oceanogr.*, **28**, 805–830.
- Habata, S., M. Yokokawa, and S. Kitawaki, 2003: The Earth Simulator System. *NEC Res. Dev.*, **44**, 21–26.
- Hall, M. M., and H. L. Bryden, 1982: Direct estimates and mechanisms of ocean heat transport. *Deep-Sea Res.*, **29A**, 339–359.
- Hogg, N. G., G. Siedler, and W. Zenk, 1999: Circulation and variability at the southern boundary of the Brazil Basin. *J. Phys. Oceanogr.*, **29**, 145–157.
- Hughes, T. M. C., and A. J. Weaver, 1994: Multiple equilibria of

- an asymmetric two-basin model. *J. Phys. Oceanogr.*, **24**, 619–637.
- Hunke, E. C., and J. K. Dukowicz, 1997: An elastic–viscous–plastic model for sea ice dynamics. *J. Phys. Oceanogr.*, **27**, 1849–1867.
- , and W. H. Lipscomb, 2004: CICE: The Los Alamos Sea Ice Model, documentation and software, Version 3.1, LA-CC-98-16. Los Alamos National Laboratory, Los Alamos, NM, 56 pp.
- Johns, T. C., and Coauthors, 2003: Anthropogenic climate change for 1860 to 2100 simulated with the HadCM3 model under updated emissions scenarios. *Climate Dyn.*, **20**, doi:10.1007/s00382-002-0296-y.
- , and Coauthors, 2004: HadGEM1—Model description and analysis of preliminary experiments for the IPCC Fourth Assessment Report. Hadley Centre Tech. Note 55, Met Office, Exeter, United Kingdom. [Available online at www.metoffice.gov.uk/research/hadleycentre/pubs/HCTN/index.html.]
- Jones, A., D. L. Roberts, M. J. Woodage, and C. E. Johnson, 2001: Indirect sulphate aerosol forcing in a climate model with an interactive sulphur cycle. *J. Geophys. Res.*, **106**, 20 293–20 310.
- Josey, S. A., E. C. Kent, D. Oakley, and P. K. Taylor, 1996: A new global air–sea heat and momentum climatology. *International WOCE Newsletter*, No. 24, WOCE International Project Office, Southampton, United Kingdom, 3–5.
- Kraus, E. B., and J. S. Turner, 1967: A one dimensional model of the seasonal thermocline II. The general theory and its consequences. *Tellus*, **19**, 98–105.
- Large, W. G., J. C. McWilliams, and S. C. Doney, 1994: Oceanic vertical mixing: A review and a model with a nonlocal boundary layer parameterization. *Rev. Geophys.*, **32**, 363–403.
- Levitus, S., and Coauthors, 1998: *World Ocean Database 1998*. Vol. 1, NOAA Atlas NESDIS 18, 346 pp.
- Lipscomb, W. H., 2001: Remapping the thickness distribution in sea ice models. *J. Geophys. Res.*, **106**, 13 989–14 000.
- Lock, A. P., A. R. Brown, M. R. Bush, G. M. Martin, and R. N. B. Smith, 2000: A new boundary layer mixing scheme. Part I: Scheme description and SCM tests. *Mon. Wea. Rev.*, **128**, 3187–3199.
- Lu, R., B. Dong, R. D. Cess, and G. L. Potter, 2004: The 1997/98 El Niño: A test for climate models. *Geophys. Res. Lett.*, **31**, L12216, doi:10.1029/2004GL019956.
- Martin, G. M., M. A. Ringer, V. D. Pope, A. Jones, C. Dearden, and T. J. Hinton, 2006: The physical properties of the atmosphere in the new Hadley Centre Global Environmental Model (HadGEM1). Part I: Model description and global climatology. *J. Climate*, **19**, 1274–1301.
- Murphy, J. M., D. M. H. Sexton, D. N. Barnett, G. S. Jones, M. J. Webb, M. Collins, and D. A. Stainforth, 2004: Quantification of modelling uncertainties in a large ensemble of climate change simulations. *Nature*, **430**, 768–772.
- Neelin, J. D., D. S. Battisti, A. C. Hirst, F.-F. Jin, Y. Wakata, T. Yamagata, and S. E. Zebiak, 1998: ENSO theory. *J. Geophys. Res.*, **103** (C7), 14 261–14 290.
- Norton, W. A., 2004: HiGEM1 on HPCx: High resolution global environmental modelling. *HPCx Capability Computing Newsletter*, No. 3, HPCx Capability Computing, 6–7. [Available online at www.hpcx.ac.uk/about/newsletter.]
- Oki, T., and Y. C. Sud, 1998: Design of Total Runoff Integrating Pathways (TRIP)—A global river channel network. *Earth Interactions*, **2**. [Available online at <http://EarthInteractions.org>.]
- Pacanowski, R. C., and S. M. Griffies, 1998: MOM 3.0 manual. NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, NJ, 692 pp. [Available from NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, NJ 08542.]
- Pardaens, A. K., H. T. Banks, J. M. Gregory, and P. R. Rowntree, 2003: Freshwater transports in HadCM3. *Climate Dyn.*, **21**, 177–195.
- Peters, H., M. C. Gregg, and J. M. Toole, 1988: On the parameterization of equatorial turbulence. *J. Geophys. Res.*, **93**, 1199–1218.
- Pope, V. D., M. L. Gallani, P. R. Rowntree, and R. A. Stratton, 2000: The impact of new physical parametrizations in the Hadley Centre climate model: HadAM3. *Climate Dyn.*, **16**, 123–146.
- Ramaswamy, V., and Coauthors, 2001: Radiative forcing of climate change. *Climate Change 2001: The Scientific Basis*, J. T. Houghton et al., Eds., Cambridge University Press, 349–416.
- Rayner, N. A., D. E. Parker, E. B. Horton, C. K. Folland, L. V. Alexander, D. P. Rowell, E. C. Kent, and A. Kaplan, 2003: Global analyses of sea surface temperature, sea ice, and night marine air temperature since the late nineteenth century. *J. Geophys. Res.*, **108**, 4407, doi:10.1029/2002JD002670.
- Ringer, M. A., and Coauthors, 2006: The physical properties of the atmosphere in the new Hadley Centre Global Environmental Model (HadGEM1). Part II: Aspects of variability and regional climate. *J. Climate*, **19**, 1302–1326.
- Roberts, D. L., and A. Jones, 2004: Climate sensitivity to black carbon aerosol from fossil fuel combustion. *J. Geophys. Res.*, **109**, D16202, doi:10.1029/2004JD004676.
- Roberts, M. J., 2004: The Gent and McWilliams parameterisation scheme, including Visbeck and biharmonic GM schemes. Tech. Rep., Unified Model Documentation Paper UMDP54, Met Office, 23 pp.
- , and D. Marshall, 1998: Do we require adiabatic dissipation schemes in eddy-resolving ocean models? *J. Phys. Oceanogr.*, **28**, 2050–2063.
- , and Coauthors, 2004: Impact of an eddy-permitting ocean resolution on control and climate change simulations with a global coupled GCM. *J. Climate*, **17**, 3–20.
- Roether, W., V. M. Roussinov, and R. Well, 1994: A tracer study of the thermohaline circulation of the eastern Mediterranean. *Ocean Processes in Climate Dynamics: Global and Mediterranean Examples*, P. Malanotte-Rizzoli and A. R. Robinson, Eds., Kluwer Academic Press, 371–394.
- Saji, N. H., B. N. Goswami, P. N. Vinayachandran, and T. Yamagata, 1999: A dipole mode in the tropical Indian Ocean. *Nature*, **401**, 360–363.
- Sato, T., 2004: The current status of the Earth Simulator. *J. Earth Simulator*, **1**, 6–7.
- Semtner, A. J., 1976: A model for the thermodynamic growth of sea ice in numerical investigations of climate. *J. Phys. Oceanogr.*, **6**, 379–389.
- Smith, W. H. F., and D. Sandwell, 1997: Global seafloor topography from satellite altimetry and ship depth soundings. *Science*, **277**, 1956–1962.
- Spencer, H., R. T. Sutton, J. M. Slingo, M. J. Roberts, and E. Black, 2005: The Indian Ocean climate and dipole variability in the Hadley Centre coupled GCMs. *J. Climate*, **18**, 2286–2307.
- Stott, P. A., G. S. Jones, J. A. Lowe, P. W. Thorne, C. F. Durman, T. C. Johns, and J.-C. Thelen, 2006: Transient climate simu-

- lations with the HadGEM1 climate model: Causes of past warming and future climate change. *J. Climate*, in press.
- Stouffer, R. J., A. J. Weaver, and M. Eby, 2004: A method for obtaining pre-twentieth century initial conditions for use in climate change studies. *Climate Dyn.*, **23**, 327–339.
- Suarez, M. J., and P. S. Schopf, 1988: A delayed oscillator for ENSO. *J. Atmos. Sci.*, **45**, 3283–3288.
- Taylor, K. E., 2001: Summarizing multiple aspects of model performance in a single diagram. *J. Geophys. Res.*, **106** (D7), 7183–7192.
- Thompson, S. R., 1995: Sills of the global ocean: A compilation. *Ocean Modell.*, **109**, 7–9.
- Thorndike, A. S., D. A. Rothrock, G. A. Maykut, and R. Colony, 1975: The thickness distribution of sea ice. *J. Geophys. Res.*, **80**, 4501–4513.
- Turner, A. G., P. M. Inness, and J. M. Slingo, 2005: The role of the basic state in the ENSO–monsoon relationship and implications for predictability. *Quart. J. Roy. Meteor. Soc.*, **131**, 781–804.
- Visbeck, M., J. Marshall, T. Haine, and M. Spall, 1997: Specification of eddy transfer coefficients in coarse-resolution ocean circulation models. *J. Phys. Oceanogr.*, **27**, 381–402.
- Webb, M. J., C. A. Senior, S. Bony, and J.-J. Morcrette, 2001: Combining ERBE and ISCCP data to assess clouds in the Hadley Centre, ECMWF and LMD atmospheric climate models. *Climate Dyn.*, **17**, 905–922.
- , and Coauthors, 2006: On the contribution of local feedback mechanisms to the range of climate sensitivity in two GCM ensembles. *Climate Dyn.*, in press.
- Webster, P. J., A. M. Moore, J. P. Loschnigg, and R. R. Leben, 1999: Coupled ocean–atmosphere dynamics in the Indian Ocean during 1997–98. *Nature*, **401**, 356–360.
- Webster, S., A. R. Brown, D. R. Cameron, and C. P. Jones, 2003: Improvements to the representation of orography in the Met Office Unified Model. *Quart. J. Roy. Meteor. Soc.*, **129**, 1989–2010.
- Wijffels, S., F. Firing, and H. Bryden, 1994: Direct observations of the Ekman balance at 10°N in the Pacific. *J. Phys. Oceanogr.*, **24**, 1666–1679.
- Williams, K. D., M. A. Ringer, and C. A. Senior, 2003: Evaluating the cloud response to climate change and current climate variability. *Climate Dyn.*, **20**, 705–721.
- Wilson, D. R., and S. P. Ballard, 1999: A microphysically based precipitation scheme for the Meteorological Office Unified Model. *Quart. J. Roy. Meteor. Soc.*, **125**, 1607–1636.
- Woodage, M., P. Davison, and D. L. Roberts, 2003: Aerosol processes. Unified Model Documentation Paper 20, Met Office, Exeter, United Kingdom, 30 pp.
- Woolnough, S. J., J. M. Slingo, and B. J. Hoskins, 2000: The relationship between convection and sea surface temperature on intraseasonal timescales. *J. Climate*, **13**, 2086–2104.
- Xie, P., and P. A. Arkin, 1997: Global precipitation: A 17-year monthly analysis based on gauge observations, satellite estimates and numerical model outputs. *Bull. Amer. Meteor. Soc.*, **78**, 2539–2558.
- Yamagata, T., S. K. Behera, J. J. Luo, S. Masson, M. R. Jury, and S. A. Rao, 2004: Coupled ocean–atmosphere variability in the tropical Indian Ocean. *Earth's Climate: The Ocean–Atmosphere Interaction*, *Geophys. Monogr.*, No. 147, Amer. Geophys. Union, 189–211.
- Zebiak, S. E., and M. A. Cane, 1987: A model El Niño–Southern Oscillation. *Mon. Wea. Rev.*, **115**, 2262–2278.
- Zenk, W., G. Siedler, B. Lenz, and N. G. Hogg, 1999: Antarctic Bottom Water flow through the Hunter Channel. *J. Phys. Oceanogr.*, **29**, 2785–2801.
- Zhang, Y.-C., W. B. Rossow, A. A. Lacis, V. Oinas, and M. I. Mishchenko, 2004: Calculation of radiative flux profiles from the surface to top of atmosphere based on ISCCP and other global data sets: Refinements of the radiative transfer model and the input data. *J. Geophys. Res.*, **109**, D19105, doi:10.1029/2003JD004457.